Online Appendix for "Expectations and Frictions: Lessons from a Quantitative Dispersed Information Model"

Rafael Gonçalves*
Marcel Ribeiro†

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*Princeton University. Email: rgoncalves@princeton.edu
†Sao Paulo School of Economics – FGV. Email: marcel.ribeiro@fgv.br

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This document provides supplementary material for the main paper. It includes the log-linearized model equations (Section A), the signal extraction problem and steady-state Kalman gain derivation (Section B), a detailed description of the data construction and estimation details including measurement equations (Section C), the solution method proofs and algorithm (Section D), posterior estimates from robustness exercises (Section E), the higher-order truncation analysis (Section F), the detrending procedure and full derivation of the optimality conditions (Sections G, H, and I), and impulse response functions (Section J). Please refer to the main paper for details on the model.

A Log-linearized model

The log-linearization of the model's optimality conditions leads to the following system of equations. For the derivation of these equations from first-order conditions, including the

detrending procedure, see Sections G–I.

$$c_t = \frac{\varphi/\gamma}{1+\varphi/\gamma} c_{t-1} + \frac{1}{1+\varphi/\gamma} \bar{E}_t[c_{t+1}] - \frac{1-\varphi/\gamma}{1+\varphi/\gamma} \bar{E}_t[r_t - \pi_{t+1} - (x^c_{t+1} - x^c_t)] \tag{A.1}$$

$$i_t = \left(\frac{1}{1+\beta}\right) i_{t-1} + \left(\frac{\beta}{1+\beta}\right) \bar{E}_t[i_{t+1}] + \left(\frac{1}{s''(1+\beta)\gamma^2}\right) \left(q_t + \bar{E}_t\left[x^i_t\right]\right) \tag{A.2}$$

$$q_t = \left(\frac{\beta(1-\delta)}{\gamma}\right) \bar{E}_t[q_{t+1}] + \left(1 - \frac{\beta(1-\delta)}{\gamma}\right) \bar{E}_t\left[r^k_{t+1}\right] - \bar{E}_t[r_t - \pi_{t+1}] \tag{A.3}$$

$$k_t = \left(\frac{1-\delta}{\gamma}\right) k_{t-1} + \left(1 - \frac{1-\delta}{\gamma}\right) \left(i_t + x^i_t\right) \tag{A.4}$$

$$u_t = \left(\bar{R}^k/a''\right) \bar{E}_t[r^k_t] \tag{A.5}$$

$$k^u_t = k_{t-1} + u_t \tag{A.6}$$

$$mrs_t = \left(\frac{1}{1-\varphi/\gamma}\right) (c_t - \varphi/\gamma c_{t-1}) + \chi l_t \tag{A.7}$$

$$y_t = \left(1 + \frac{\Phi_p}{\bar{Y}}\right) (\alpha k^u_t + (1-\alpha) l_t + x^a_t) \tag{A.8}$$

$$k^u_t - l_t = w_t - r^k_t \tag{A.9}$$

$$mc_t = (1-\alpha) w_t + \alpha r^k_t - x^a_t \tag{A.10}$$

$$y_t = \frac{1}{1-g_y} \left(\frac{\bar{C}}{\bar{Y}} c_t + \frac{\bar{I}}{\bar{Y}} i_t + \frac{\bar{R}^k \bar{K}}{\bar{Y}} u_t + x^g_t\right) \tag{A.11}$$

$$r_t = \phi_r r_{t-1} + (1-\phi_r)(\phi_\pi \pi_t + \phi_y y_t) + x^r_t \tag{A.12}$$

$$\pi_t = \xi_p \iota_p \pi_{t-1} + \kappa_p \xi_p \bar{E}_t[mc_t + x^p_t] + (1-\xi_p)(1-\iota_p \beta \xi_p) \bar{E}_t[\pi_t] + (1-\xi_p)\beta\xi_p \int_0^1 E_{i,t}[p^*_{i,t+1}] di \tag{A.13}$$

$$w_t + \pi_t = \xi_w(\iota_w \pi_{t-1} + w_{t-1}) + (1-\xi_w) w^*_t \tag{A.14}$$

$$w^*_t = \frac{(1-\beta\xi_w)}{1+\chi\theta_w} \bar{E}_t[mrs_t + \chi\theta_w w_t + x^w_t] + (1-\iota_w\beta\xi_w)\bar{E}_t[\pi_t] + \beta\xi_w \int_0^1 E_{h,t}\left[w^*_{h,t+1}\right] dh \tag{A.15}$$

where $\kappa_p = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}$, $\kappa_w = \frac{(1-\xi_w)(1-\beta\xi_w)}{\xi_w(1+\chi\theta_w)}$ and $\theta_w = \frac{1+\bar{\mu}^w}{\bar{\mu}^w}$. $\bar{E}_t[\cdot] \equiv \int_0^1 E_{i,t}[\cdot] di$ denotes average expectation, where $E_{i,t}[\cdot] \equiv E[\cdot \mid \mathcal{I}_{it}]$ is agent i's expectation based on her information set at period t, $\mathcal{I}_{it}$.

All equations are standard in macroeconomic models. Equation (A.1) is the Euler equation for consumption. Equation (A.2) is the equilibrium condition for investment, and Equation (A.3) determines the marginal real value of a unit of capital q_t. Equation (A.4) is the law of motion of capital. Equation (A.5) is the optimal condition for capital utilization,

and Equation (A.6) comes from the definition of utilized capital. Equations (A.7) and (A.8) are the marginal rate of substitution between consumption and labor and the production function, respectively. Equations (A.9) and (A.10) come from firms' cost minimization. Equation (A.11) derives from the aggregate resource constraint, while Equation (A.12) is the Taylor rule. Finally, Equation (A.13) is the New Keynesian Phillips curve for prices, Equation (A.14) describes wage dynamics, and Equation (A.15) is the optimal wage setting condition.

B Signal extraction and steady-state Kalman gain

In section 4 of the main paper, we estimate the steady-state Kalman gain for each shock directly. While the Kalman gain is an endogenous object, our information structure implies that it has a one-to-one relationship to the private signal variance, conditional on the shock persistence and variance. In the following, we show that this is the case.

Recall that all shocks have the same AR(1) structure given by

$$x_t^l = \rho_l x_{t-1}^l + \varepsilon_t^l, \quad \varepsilon_t^l \sim \mathcal{N}(0, \sigma_l^2). \tag{B.1}$$

where $l \in \{a, c, i, g, p, w, r\}$.

Agents use information from their signals, restated here for convenience:

$$s_{j,t}^l = x_t^l + v_{j,t}^l, \quad v_{j,t}^l \sim \mathcal{N}(0, \tau_l^2). \tag{B.2}$$

Agents must form expectations about the whole hierarchy of expectations of all shocks and the solution method takes this into account. Here we simply show how our measure of information friction – the steady-state Kalman gain from the first-order expectation of individual j – can be found as a function of deep parameters.

Since shocks and their respective signals are independent, their signal extraction can be written separately. The rational expectation of each shock l can be computed by the Kalman filter using (B.1) as state equation and (B.2) as measurement equation.

The update equation of the Kalman filter is given by

$$E_{j,t}[x_t^l] = E_{j,t-1}[x_t^l] + \bar{k}_l \left[s_{j,t}^l - E_{j,t-1}[s_{j,t}^l] \right] \tag{B.3}$$

where $\bar{k}_l$ is the steady-state Kalman gain. The steady-state Kalman gain for each shock l can be computed by the standard Kalman filter equations applied to the state equation (B.1) and measurement equation (B.2) and evaluated at the steady-state. This implies the

following expressions:

$$\bar{k}_l = \frac{\bar{p}_l}{\bar{p}_l + \tau_l^2} \tag{B.4}$$

$$\bar{p}_l = \rho_l^2(1 - \bar{k}_l)\bar{p}_l + \sigma_l^2 \tag{B.5}$$

where $\bar{k}_l$ and $\bar{p}_l$ are the Kalman gain and the mean squared error (MSE) of the one-period ahead prediction error both at steady-state.

Note that the Kalman gain can be expressed alternatively in terms of the MSE relative to the shock variance such that

$$\bar{k}_l = \frac{\tilde{p}_l}{\tilde{p}_l + r_l^2}$$

$$\tilde{p}_l = \rho_l^2(1 - \bar{k}_l)\tilde{p}_l + 1$$

where $\tilde{p}_l = \bar{p}_l/\sigma_l^2$ is the relative MSE and $r_l = \tau_l/\sigma_l$ the noise-to-signal ratio. The first equation simply divides the denominator and numerator of equation (B.4) by σ_l^2 while the second equation can be found by dividing equation (B.5) by σ_l^2 on both sides.

Both equations lead to the quadratic Riccati equation in terms of the noise-to-signal ratio:

$$\tilde{p}_l^2 - \left(1 - (1 - \rho_l^2)r_l^2\right)\tilde{p}_l - r_l^2 = 0 \tag{B.6}$$

This equation also has a unique positive solution, leading to $\tilde{p}_l = \tilde{p}_l(\rho_l, r_l)$.

This implies that the steady-state Kalman filter depends only on the persistence of the shock and the noise-to-signal ratio $\bar{k}_l = \bar{k}_l(\rho_l, r_l) \in (0, 1)$.

C Data

This appendix describes the construction of the observed macroeconomic variables and forecast revisions used in the estimation. Data on macroeconomic aggregates are collected from the Federal Reserve Database (FRED) and expectation data come from the Survey of Professional Forecasters (SPF). The time series collected to construct the database are displayed in Table C1. The data starts in 1981Q4 due to the expectation data collection and ends in 2007Q4, before the Great Recession, to avoid issues with lower zero bound of the interest rate.

All variables from the Survey of Professional Forecasters (SPF) are suffixed with a number indicating the forecast horizon. We use horizons 1 through 3, which correspond to the

Table C1: Database

Series	Description	Measure	Source
GDPC1	Real Gross Domestic Product	Billions of Chained 2012 Dollars	BEA
PCECC96	Real Personal Consumption Expenditures	Billions of Chained 2012 Dollars	BEA
GPDIC1	Real Gross Private Domestic Investment	Billions of Chained 2012 Dollars	BEA
GDPDEF	Gross Domestic Product: Implicit Price Deflator	Index 2012=100	BEA
DTB3	3-Month Treasury Bill Secondary Market Rate	Percentage points	FED Board
PRS85006063	Nonfarm Business Sector: Compensation	Index 2012=100	BEA
HOANBS	Nonfarm Business Sector: Hours of All Persons	Index 2012=100	BLS
CNP16OV	Civilian Noninstitutional Population	Thousands of People	BLS
RGDP	Average forecast for the Real Gross Domestic Product	Chained Dollars (base year varies)	SPF
RCONSUM	Average forecast for the Real Personal Consumption Expenditures	Chained Dollars (base year varies)	SPF
RNRESINV	Average forecast for the Real Nonresidential Fixed Investment	Chained Dollars (base year varies)	SPF
RRESINV	Average forecast for the Real Residential Fixed Investment	Chained Dollars (base year varies)	SPF
RCBI	Average forecast for the Real change in Private Inventories	Chained Dollars (base year varies)	SPF
PGDP	Average forecast for the GDP Price Index	Index (base year varies)	SPF
TBILL	Average forecast for the annualized three-month Treasury bill rate	Percentage points	SPF

Note: BEA, BLS, Fed Board and SPF stand for, respectively: U.S. Bureau of Economic Analysis, U.S. Bureau of Labor Statistics, Board of Governors of the Federal Reserve and Survey of Professional Forecasters.

previous quarter, the current quarter, and one-quarter ahead, respectively. For example, RGDP1 is the nowcast of real GDP for the quarter prior to the survey's release, RGDP2 is the nowcast for the current quarter, and RGDP3 is the one-quarter ahead forecast.[1]

Real output (GDPC1), consumption (PCE) and investment (GDPIC1) are calculated by dividing them by the Hodrick-Prescott (HP) filtered trend of the population index (CNP16OV), denoted by HP(CNP16OV). This method, as discussed by Edge and Gürkaynak [2010], avoids incorporating fluctuations caused by measurement changes from the Census Bureau.

Real wage is constructed by dividing the compensation in the nonfarm business sector (PRS85006063) by the GDP deflator (GDPDEF). Per capita total hours is computed by dividing hours of all persons in the nonfarm business sector (HOANBS) by the smoothed population index. Inflation is defined as the annualized quarterly growth rate of GDP deflator (GDPDEF). The nominal interest rate is measured by 3-month Treasury bill rate to be consistent with the SPF expectation data, as it does not survey forecasts for the federal funds rate.

[1] Although the Bureau of Economic Analysis's (BEA) advance estimates for the previous quarter are available, forecasters can still "predict" this quarter's data by forecasting revisions to the official data. According to the survey, most forecasters use the BEA's advanced estimate.

Therefore, the macroeconomic data in levels is constructed according to:

$$
\begin{aligned}
Y_t^{obs} &= GDPC1_t/HP(CNP16OV_t) \\
C_t^{obs} &= PCE_t/HP(CNP16OV_t) \\
I_t^{obs} &= GPDIC1_t/HP(CNP16OV_t) \\
W_t^{obs} &= PRS85006063_t/GDPDEF_t \\
L_t^{obs} &= HOANBS_t/HP(CNP16OV_t),
\end{aligned}
$$

The macroeconomic data used in the estimation is given by:

$$
\begin{aligned}
dy_t^{obs} &= 100\log(Y_t^{obs}/Y_{t-1}^{obs}) \\
dc_t^{obs} &= 100\log(C_t^{obs}/C_{t-1}^{obs}) \\
di_t^{obs} &= 100\log(I_t^{obs}/I_{t-1}^{obs}) \\
dw_t^{obs} &= 100\log(W_t^{obs}/W_{t-1}^{obs}) \\
l_t^{obs} &= L_t^{obs} - \bar{L}^{obs} \\
\pi_t^{obs} &= 400\log(GDPDEF_t/GDPDEF_{t-1}) \\
r_t^{obs} &= DTB3_t,
\end{aligned}
$$

where $\bar{L}^{obs}$ denotes the sample mean of total hours per capita, L_t^{obs}.

Now we turn to the expectation data. Investment expectations in the SPF are broken down into three components: nonresidential fixed investment (RNRESIN), residential fixed investment (RRESINV), and change in private inventories (RCBI). Thus, we construct forecasts for Real Gross Private Domestic Investment, $RINVEST_t = RNRESIN_t + RRESINV_t + RCBI_t$, to align with the GPDIC1 index.

It is important to note that the SPF's expectation data is in chained dollars with varying base years, while the actual data is chained to 2012 dollars. Additionally, because the SPF does not survey expectations for the population index, we construct per capita forecast variables using the same index as the real variables. Thus, our measure of expectation revisions on per capita variables is not affected by population growth.

Thus, the expectation data in levels and first differences are given by:

$$
\begin{aligned}
\bar{E}_t^{obs}[Y_{t+1}] &= RGDP3_t/HP(CNP16OV_{t+1}) \\
\bar{E}_t^{obs}[Y_t] &= RGDP2_t/HP(CNP16OV_t) \\
\bar{E}_t^{obs}[Y_{t-1}] &= RGDP1_t/HP(CNP16OV_{t-1}) \\
\bar{E}_t^{obs}[C_{t+1}] &= RCONSUM3_t/HP(CNP16OV_{t+1}) \\
\bar{E}_t^{obs}[C_t] &= RCONSUM2_t/HP(CNP16OV_t) \\
\bar{E}_t^{obs}[C_{t-1}] &= RCONSUM1_t/HP(CNP16OV_{t-1}) \\
\bar{E}_t^{obs}[I_{t+1}] &= RINVEST3_t/HP(CNP16OV_{t+1}) \\
\bar{E}_t^{obs}[I_t] &= RINVEST2_t/HP(CNP16OV_t) \\
\bar{E}_t^{obs}[I_{t-1}] &= RINVEST1_t/HP(CNP16OV_{t-1}) \\
\bar{E}_t^{obs}[dy_{t+1}] &= 100\log\left(\bar{E}_t^{obs}[Y_{t+1}]/\bar{E}_t^{obs}[Y_t]\right) \\
\bar{E}_t^{obs}[dc_{t+1}] &= 100\log\left(\bar{E}_t^{obs}[C_{t+1}]/\bar{E}_t^{obs}[C_t]\right) \\
\bar{E}_t^{obs}[di_{t+1}] &= 100\log\left(\bar{E}_t^{obs}[I_{t+1}]/\bar{E}_t^{obs}[I_t]\right) \\
\bar{E}_t^{obs}[dy_t] &= 100\log\left(\bar{E}_t^{obs}[Y_t]/\bar{E}_t^{obs}[Y_{t-1}]\right) \\
\bar{E}_t^{obs}[dc_t] &= 100\log\left(\bar{E}_t^{obs}[C_t]/\bar{E}_t^{obs}[C_{t-1}]\right) \\
\bar{E}_t^{obs}[di_t] &= 100\log\left(\bar{E}_t^{obs}[I_t]/\bar{E}_t^{obs}[I_{t-1}]\right) \\
\bar{E}_t^{obs}[\pi_{t+1}] &= 400\log(PGDP3_t/PGDP2_t) \\
\bar{E}_t^{obs}[\pi_t] &= 400\log(PGDP2_t/PGDP1_t) \\
\bar{E}_t^{obs}[r_{t+1}] &= TBILL3_t \\
\bar{E}_t^{obs}[r_t] &= TBILL2_t
\end{aligned}
$$

where $\bar{E}_s^{obs}[X_t]$ denotes data on the average expectation about X on horizon t based on information of period s. Growth rates are expressed as percentages, while inflation and the nominal interest rate are in annualized percentage points.

Our primary expectation series is the average current forecast revision, defined as the difference between the nowcast of a variable for the current quarter and the previous quarter's forecast for that same variable. Thus, the expectation data used in the estimation is given

by:

$$
\begin{aligned}
\overline{Rev}^{obs}_{t|t-1}[dy_t] &= \bar{E}^{obs}_t[dy_t] - \bar{E}^{obs}_{t-1}[dy_t] \\
\overline{Rev}^{obs}_{t|t-1}[dc_t] &= \bar{E}^{obs}_t[dc_t] - \bar{E}^{obs}_{t-1}[dc_t] \\
\overline{Rev}^{obs}_{t|t-1}[di_t] &= \bar{E}^{obs}_t[di_t] - \bar{E}^{obs}_{t-1}[di_t] \\
\overline{Rev}^{obs}_{t|t-1}[\pi_t] &= \bar{E}^{obs}_t[\pi_t] - \bar{E}^{obs}_{t-1}[\pi_t] \\
\overline{Rev}^{obs}_{t|t-1}[r_t] &= \bar{E}^{obs}_t[r_t] - \bar{E}^{obs}_{t-1}[r_t]
\end{aligned}
$$

$\overline{Rev}^{obs}_{k|s}[X_t] = \bar{E}^{obs}_s[X_t] - \bar{E}^{obs}_{s-1}[X_t]$ denotes the revision of the expectation about the same variable X on horizon t based on new information from period k relative to period s.

C.1 Estimation

The log-linearized equilibrium conditions are reported in Section A, while the data construction is detailed in Section C. Below we present the measurement equations linking observables to model variables.

C.1.1 Measurement equations

The following measurement equations link the observed data to the model's variables:

$$
\begin{aligned}
dy^{obs}_t &= \bar{\gamma} + y_t - y_{t-1} \\
dc^{obs}_t &= \bar{\gamma} + c_t - c_{t-1} \\
di^{obs}_t &= \bar{\gamma} + i_t - i_{t-1} \\
dw^{obs}_t &= \bar{\gamma} + w_t - w_{t-1} \\
l^{obs}_t &= l_t \\
\pi^{obs}_t &= \bar{\pi} + 4\pi_t \\
r^{obs}_t &= \bar{r} + 4r_t \\
\overline{Rev}^{obs}_{t|t-1}[dy_t] &= \bar{E}_t[\Delta y_t] - \bar{E}_{t-1}[\Delta y_t] + \varepsilon^{me}_{dy,t} \\
\overline{Rev}^{obs}_{t|t-1}[dc_t] &= \bar{E}_t[\Delta c_t] - \bar{E}_{t-1}[\Delta c_t] + \varepsilon^{me}_{dc,t} \\
\overline{Rev}^{obs}_{t|t-1}[di_t] &= \bar{E}_t[\Delta i_t] - \bar{E}_{t-1}[\Delta i_t] + \varepsilon^{me}_{di,t} \\
\overline{Rev}^{obs}_{t|t-1}[\pi_t] &= 4\left(\bar{E}_t[\pi_t] - \bar{E}_{t-1}[\pi_t]\right) + \varepsilon^{me}_{\pi,t} \\
\overline{Rev}^{obs}_{t|t-1}[r_t] &= 4\left(\bar{E}_t[r_t] - \bar{E}_{t-1}[r_t]\right) + \varepsilon^{me}_{r,t},
\end{aligned}
\tag{C.1}
$$

where $\bar{\gamma} \equiv 100(\gamma - 1)$ is the quarterly trend growth rate of productivity. The variables with superscript "*obs*" correspond to the variables in the database. The notation dx_t^{obs} denotes the quarterly growth rate of the variables $x \in \{y, c, i, w\}$ at period t. The data on hours worked (l_t^{obs}) is demeaned, while inflation (π_t^{obs}) and nominal interest rate (r_t^{obs}) are computed in annual terms. $\bar{\pi} \equiv 400(\bar{\Pi} - 1)$ and $\bar{r} \equiv 400(\bar{R} - 1)$ stand for the steady-state annualized inflation and nominal interest rates, respectively.

The variables $\overline{Rev}_{t|t-1}^{obs}[x_t]$, for $x \in \{dy, dc, di, \pi, r\}$, are the current forecast revision. Measurement errors ε_x^{me} in the observational equations associated with forecast revisions follow an i.i.d. Gaussian distribution with standard deviations denoted by σ_x^{me}, with $x \in \{dy, dc, di, \pi, r\}$.

D Solution method and Algorithm

We first start with two Lemmas. The first writes the first-order expectation of the hierarchy as a linear transformation of the hierarchy.

Lemma 1. *The first-order expectation of the hierarchy of expectations satisfy*

$$E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] = T x_t^{(0:\bar{k})} \tag{D.1}$$

where $T = \begin{bmatrix} 0_{n\bar{k}\times n} & I_{n\bar{k}} \\ 0_{n\times n} & 0_{n\times n\bar{k}} \end{bmatrix}$ *is the order transformation matrix.*

Proof. By the definition of $x_t^{(0:\bar{k})}$ one can see that $E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] = x_t^{(1:\bar{k}+1)}$. By definition for $\bar{k}$, any order s such that $s > \bar{k}$ does not affect the equilibrium. Then, without loss of generality, one can set $E_t^{(s)}[x_t] = 0$ if $s > \bar{k}$. Therefore, one can rewrite $E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]$ as

$$E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] = \begin{bmatrix} x_t^{(1:\bar{k})} \\ E_t^{(\bar{k}+1)}[x_t] \end{bmatrix} = \begin{bmatrix} x_t^{(1:\bar{k})} \\ 0_{n\times 1} \end{bmatrix} = \begin{bmatrix} 0_{n\bar{k}\times n} & I_{n\bar{k}} \\ 0_{n\times n} & 0_{n\times n\bar{k}} \end{bmatrix} \begin{bmatrix} x_t \\ x_t^{(1:\bar{k})} \end{bmatrix} = T x_t^{(0:\bar{k})}, \tag{D.2}$$

where the first equality uses the definition of $x_t^{(1:\bar{k}+1)}$ and the second equality uses that $E_t^{(\bar{k}+1)}[x_t] = 0$. In the last equality, T is defined accordingly.

□

This Lemma builds on Proposition 1 from Online Appendix of Melosi [2017]. It explores the truncation of the hierarchy.

The second Lemma does the opposite: writes the hierarchy as a linear transformation of the first-order expectation of the hierarchy.

Lemma 2. *$x_t^{(0:\bar{k})}$ can be rewritten as a linear function of x_t and its average expectation such that*

$$x_t^{(0:\bar{k})} \equiv e_x' x_t + T' E_t^{(1)} \left[x_t^{(0:\bar{k})} \right],$$

Proof. By definition, $x_t^{(0:\bar{k})}$ can be decomposed as $x_t^{(0:\bar{k})} = \left[x_t' \quad (x_t^{(\bar{1}:\bar{k})})' \right]'$. Using this decomposition, the average expectation decomposed as $E_t^{(1)} \left[x_t^{(0:\bar{k})} \right] = \left[(x_t^{(\bar{1}:\bar{k})})' \quad E_t^{(\bar{k}+1)} [x_t]' \right]'$. Therefore, one can use this two definitions such that

$$x_t^{(0:\bar{k})} \equiv \begin{bmatrix} x_t \\ x_t^{(\bar{1}:\bar{k})} \end{bmatrix} = \begin{bmatrix} I_n \\ 0_{n\bar{k}\times n} \end{bmatrix} x_t + \begin{bmatrix} 0_{n\times n\bar{k}} & 0_{n\times n} \\ I_{n\bar{k}} & 0_{n\bar{k}\times n} \end{bmatrix} \begin{bmatrix} x_t^{(1:\bar{k})} \\ E_t^{(\bar{k}+1)} [x_t] \end{bmatrix} = e_x' x_t + T' E_t^{(1)} \left[x_t^{(0:\bar{k})} \right],$$

where last equality uses the definitions of T in Lemma 1 and e_x. □

Lemmas are taken from Ribeiro [2018, chap. 3].

D.1 Proof of Proposition 1

Individual and average expectations about the hierarchy. The Kalman filter delivers the individual conditional expectation, $E_{i,t}[\cdot] \equiv E[\cdot|\mathcal{I}_t^i]$, where $\mathcal{I}_t^i = \{s_{i,\tau}, \tau \leq t\}$ is the information set of individual i in period t.

The state equation is the hierarchy of expectations that is given by the state equation (35), restated for convenience:

$$x_t^{(0:\bar{k})} = \mathbf{A} x_{t-1}^{(0:\bar{k})} + \mathbf{B} \varepsilon_t, \tag{D.3}$$

and the agent i with the observational equation (32), also restated:

$$s_{i,t} = C_x x_t + D v_{i,t}.$$

Note that $s_{i,t}$ is a signal about the shocks and not the whole hierarchy of expectations.

Let the selection matrix $e_x \equiv \begin{bmatrix} I_n & 0_{n\times n\bar{k}} \end{bmatrix}$ such that $x_t = e_x x_t^{(0:\bar{k})}$. Then, we can rewrite the signal in terms of the hierarchy as

$$s_{i,t} = C x_t^{(0:\bar{k})} + D v_{it}. \tag{D.4}$$

where $C = C_x e_x$.

Each agent i uses the Kalman filter and finds the update equation given by

$$E_{i,t} \left[x_t^{(0:\bar{k})} \right] = E_{i,t-1} \left[x_t^{(0:\bar{k})} \right] + \mathbf{K_t} \left[s_{i,t} - E_{i,t-1} [s_{i,t}] \right], \tag{D.5}$$

where $\mathbf{K_t}$ is the Kalman gain given by

$$\mathbf{K_t} = \mathbf{P_{t/t-1}} C' \left[C\mathbf{P_{t/t-1}} C' + D\Sigma_v D' \right]^{-1}. \tag{D.6}$$

As usual, the mean squared error (MSE) of the one-period ahead prediction error is given by

$$\mathbf{P_{t+1/t}} = \mathbf{A} \left[\mathbf{P_{t/t-1}} - \mathbf{K_t} \left[C\mathbf{P_{t/t-1}} \right] \right] \mathbf{A}' + \mathbf{B}\Sigma_\varepsilon \mathbf{B}'. \tag{D.7}$$

For details of this derivation, see for instance Hamilton [1995, chap. 13].

Using the observational equation, (D.4), taking expectations and inserting in (D.5) one can find:

$$E_{i,t}\left[x_t^{(0:\bar{k})}\right] = E_{i,t-1}\left[x_t^{(0:\bar{k})}\right] + \mathbf{K_t}\left[Cx_t^{(0:\bar{k})} + Dv_{it} - CE_{i,t-1}\left[x_t^{(0:\bar{k})}\right]\right] \tag{D.8}$$

Therefore, one can rewrite the equation above as

$$E_{i,t}\left[x_t^{(0:\bar{k})}\right] = \left(I_k - \mathbf{K_t}C\right) E_{i,t-1}\left[x_t^{(0:\bar{k})}\right] + \mathbf{K_t}\left[Cx_t^{(0:\bar{k})} + Dv_{it}\right] \tag{D.9}$$

where $k = n(\bar{k}+1)$. Using the fact that $E_{i,t-1}\left[x_t^{(0:\bar{k})}\right] = \mathbf{A}E_{i,t-1}\left[x_{t-1}^{(0:\bar{k})}\right]$ and substituting equation (D.3), one can find:

$$E_{i,t}\left[x_t^{(0:\bar{k})}\right] = \left(I_k - \mathbf{K_t}C\right)\mathbf{A}E_{i,t-1}\left[x_{t-1}^{(0:\bar{k})}\right] + \mathbf{K_t}C\mathbf{A}x_{t-1}^{(0:\bar{k})} + \mathbf{K_t}C\mathbf{B}\varepsilon_t + \mathbf{K_t}Dv_{it} \tag{D.10}$$

We follow the literature by focusing on the stationary equilibrium. Therefore, the expectation of each individual i in the stationary equilibrium is the one which the MSE is in steady-state, i.e., agents update their forecast based on the steady-state Kalman gain. In other words, the dynamics of expectations depends only on the properties of the process they are forecasting and signals, but does not depend on the period t.

Combining equations (D.6-D.7), one can find the Riccati equation

$$\mathbf{P_{t+1/t}} = \mathbf{A}\left[\mathbf{P_{t/t-1}} - \mathbf{P_{t/t-1}}C'\left[C\mathbf{P_{t/t-1}}C' + D\Sigma_v D'\right]^{-1} C\mathbf{P_{t/t-1}}\right]\mathbf{A}' + \mathbf{B}\Sigma_\varepsilon\mathbf{B}' \tag{D.11}$$

Therefore, one need to iterate this equation to find the steady-state MSE, $\bar{\mathbf{P}}$, and compute its counterpart Kalman gain, $\bar{\mathbf{K}}$. Nimark [2017] shows that if is x_t stationary process, then the expectations hierarchy about this process, $x_t^{(0:\bar{k})}$, is also stationary. Given this and the fact that Σ_ε is positive definite, there exists a steady-state solution such that $\bar{\mathbf{P}} = \mathbf{P_{t+1|t}} = \mathbf{P_{t|t-1}}$ which implies the steady-state Kalman gain $\bar{\mathbf{K}} = \mathbf{K_t} = \mathbf{K_{t-1}}$ [see Hamilton, 1995, chap. 13].

Expressions (37) in the Proposition 1 are equations (D.6-D.7) using the steady-state Kalman gain, $\bar{\mathbf{K}}$, instead of $\mathbf{K_t}$.

Moreover, the average expectation is easily computed by

$$\bar{E}_t\left[x_t^{(0:\bar{k})}\right] \equiv \int_0^1 E_{i,t}\left[x_t^{(0:\bar{k})}\right] di = \left(I_k - \bar{\mathbf{K}}C\right)\mathbf{A}\bar{E}_{t-1}\left[x_{t-1}^{(0:\bar{k})}\right] + \bar{\mathbf{K}}C\mathbf{A}x_{t-1}^{(0:\bar{k})} + \bar{\mathbf{K}}C\mathbf{B}\varepsilon_t \tag{D.12}$$

Analogously, the individual and average expectation of Proposition 1 are equations (D.10) and (D.12) using the steady-state Kalman gain, $\bar{\mathbf{K}}$, instead of $\mathbf{K_t}$.

Verify guess for $x_t^{(0:\bar{k})}$. In the first part of the proof, we found the average expectation (D.12) for the guessed dynamics of the hierarchy of expectations (D.3).

Now we verify the guessed hierarchy and find the coefficients $(\mathbf{A}, \mathbf{B})$ consistent with the average expectation.

Substituting the average expectation from equation (D.12) into the expression from Lemma 2 one can find that:

$$x_t^{(0:\bar{k})} = e_x' x_t + T'\left[\left(I_k - \bar{\mathbf{K}}C\right)\mathbf{A}E_{t-1}^{(1)}\left[x_{t-1}^{(0:\bar{k})}\right] + \bar{\mathbf{K}}C\mathbf{A}x_{t-1}^{(0:\bar{k})} + \bar{\mathbf{K}}C\mathbf{B}\varepsilon_t\right]$$

Then, using the shocks definition (31) and the fact that $x_t = e_x x_t^{(0:\bar{k})}$ one can rewrite equation above as

$$x_t^{(0:\bar{k})} = e_x'\left(A_1 e_x x_{t-1}^{(0:\bar{k})} + \varepsilon_t\right) + T'\left(I_k - \bar{\mathbf{K}}C\right)AE_{t-1}^{(1)}\left[x_{t-1}^{(0:\bar{k})}\right] + T'\bar{\mathbf{K}}CAx_{t-1}^{(0:\bar{k})} + T'\bar{\mathbf{K}}C\mathbf{B}\varepsilon_t$$

Using Lemma 1 at period $t-1$ into equation above and rearranging:

$$x_t^{(0:\bar{k})} = \left[e_x' A_1 e_x + T'\left(I_k - \bar{\mathbf{K}}C\right)AT + T'\bar{\mathbf{K}}CA\right]x_{t-1}^{(0:\bar{k})} + \left[T'\bar{\mathbf{K}}C\mathbf{B} + e_x'\right]\varepsilon_t. \tag{D.13}$$

This expression shows that the expectations hierarchy is a function of its lag and structural shocks, as guessed in equation (D.3). Therefore, the expression above verifies that $x_t^{(0:\bar{k})}$ follows the guessed form and the square brackets terms provide identities for A and $\mathbf{B}$ in equations (36)

□

D.2 Proof of Proposition 2

This proof builds on techniques developed by Ribeiro [2018, chap. 3]. The key difference is that we guess a law of motion for individual endogenous variables instead of guessing

the aggregate. This allows recovering expressions for solving for $(\mathbf{Q_0}, \mathbf{Q_1})$ and then get the solution for $\mathbf{Q}$. We also consider the exogenous information case only.

The guessed law of motion for individual endogenous variables is given (33), restated for convenience:

$$Y_{i,t} = \mathbf{R}Y_{i,t-1} + \mathbf{Q_0}x_t + \mathbf{Q_1}E_{i,t}\left[x_t^{(0:\bar{k})}\right]. \tag{D.14}$$

Aggregating the individual law of motion (D.14):

$$\begin{aligned} Y_t &= \mathbf{R}Y_{t-1} + \mathbf{Q_0}x_t + \mathbf{Q_1}E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] && \text{(D.15)} \\ &= \mathbf{R}Y_{t-1} + \mathbf{Q_0}x_t + \mathbf{Q_1}Tx_t^{(0:\bar{k})} && \text{(D.16)} \end{aligned}$$

where the last equality uses Lemma 1.

By computing the individual expectation of the endogenous aggregate variables, one can find that

$$\begin{aligned} E_{i,t}\left[Y_t\right] &= \mathbf{R}Y_{t-1} + \mathbf{Q_0}E_{i,t}\left[x_t\right] + \mathbf{Q_1}TE_{i,t}\left[x_t^{(0:\bar{k})}\right] \\ &= \mathbf{R}Y_{t-1} + (\mathbf{Q_0}e_x + \mathbf{Q_1}T)E_{i,t}\left[x_t^{(0:\bar{k})}\right] \\ E_t^{(1)}\left[Y_t\right] &= \mathbf{R}Y_{t-1} + (\mathbf{Q_0}e_x + \mathbf{Q_1}T)E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] \end{aligned} \tag{D.17}$$

where the first equality uses that Y_{t-1} is known, second equality uses the definition of the selection matrix, e_x, and the third equality aggregates.

Similarly, the individual expectation for endogenous variables in $t+1$:

$$\begin{aligned} E_{i,t}\left[Y_{t+1}\right] &= \mathbf{R}E_{i,t}\left[Y_t\right] + \mathbf{Q_0}E_{i,t}\left[x_{t+1}\right] + \mathbf{Q_1}TE_{i,t}\left[x_{t+1}^{(0:\bar{k})}\right] \\ &= \mathbf{R}E_{i,t}\left[Y_t\right] + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}T\mathbf{A})E_{i,t}\left[x_t^{(0:\bar{k})}\right] \\ &= \mathbf{R}^2Y_{t-1} + \left[\mathbf{R}(\mathbf{Q_0}e_x + \mathbf{Q_1}T) + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}T\mathbf{A})\right]E_{i,t}\left[x_t^{(0:\bar{k})}\right] \\ E_t^{(1)}\left[Y_{t+1}\right] &= \mathbf{R}^2Y_{t-1} + \left[\mathbf{R}(\mathbf{Q_0}e_x + \mathbf{Q_1}T) + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}T\mathbf{A})\right]E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] \end{aligned} \tag{D.18}$$

where the second equality uses equations (31) and (35), and the definition of e_x. Third equation uses equation (D.17) and the fourth aggregates.

Finally, taking the individual expectation of the law of motion (33) in period $t+1$ implies that

$$\begin{aligned} E_{i,t}[Y_{i,t+1}] &= \mathbf{R}Y_{i,t} + \mathbf{Q_0}E_{i,t}[x_{t+1}] + \mathbf{Q_1}E_{i,t}\left[E_{i,t+1}\left[x_{t+1}^{(0:\bar{k})}\right]\right] \\ &= \mathbf{R}Y_{i,t} + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})E_{i,t}\left[x_t^{(0:\bar{k})}\right] \end{aligned}$$

where the second equality uses the law of iterated expectations (which holds for i's expectation but not for the average expectation), equations (31) and (35) as before.

Aggregating the expectation above leads to

$$\begin{aligned}\int_0^1 E_{i,t}[Y_{i,t+1}]di &= \mathbf{R}Y_t + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})E_t^{(1)}[x_t^{(0:\bar{k})}] \\ &= \mathbf{R}\left[\mathbf{R}Y_{t-1} + \mathbf{Q_0}x_t + \mathbf{Q_1}E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]\right] + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})E_t^{(1)}[x_t^{(0:\bar{k})}] \\ &= \mathbf{R}^2Y_{t-1} + \mathbf{R}\mathbf{Q_0}x_t + [\mathbf{R}\mathbf{Q_1} + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})]\,E_t^{(1)}[x_t^{(0:\bar{k})}]\end{aligned} \tag{D.19}$$

where the second equality uses equation (D.16). Note that $\int_0^1 E_{i,t}[Y_{i,t+1}]di \neq E_t^{(1)}[Y_{t+1}]$.

Substituting the guessed solution (D.16) and the expectations (D.17-D.19) into the system of equations (30) one can find:

$$\begin{aligned}&F_1\left[\mathbf{R}^2Y_{t-1} + \mathbf{R}\mathbf{Q_0}x_t + [\mathbf{R}\mathbf{Q_1} + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})]\,E_t^{(1)}[x_t^{(0:\bar{k})}]\right] + \\ &F_2\left[\mathbf{R}^2Y_{t-1} + [\mathbf{R}(\mathbf{Q_0}e_x + \mathbf{Q_1}T) + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}T\mathbf{A})]\,E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]\right] + \\ &G_1\left[\mathbf{R}Y_{t-1} + \mathbf{Q_0}x_t + \mathbf{Q_1}E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]\right] + G_2\left[\mathbf{R}Y_{t-1} + (\mathbf{Q_0}e_x + \mathbf{Q_1}T)E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]\right] + \\ &HY_{t-1} + [(L_1A_1 + L_2)]\,e_xE_t^{(1)}\left[x_t^{(0:\bar{k})}\right] + M_1x_t = 0_{m\times 1}\end{aligned}$$

which can be rearranged to

$$\begin{aligned}&\left[F\mathbf{R}^2 + G\mathbf{R} + H\right]Y_{t-1} + [(F_1\mathbf{R} + G_1)\,\mathbf{Q_0} + M_1]\,x_t \\ &[F_1\,[\mathbf{R}\mathbf{Q_1} + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}\mathbf{A})] + F_2\,[\mathbf{R}(\mathbf{Q_0}e_x + \mathbf{Q_1}T) + (\mathbf{Q_0}A_1e_x + \mathbf{Q_1}T\mathbf{A})] + \\ &G_1\mathbf{Q_1} + G_2(\mathbf{Q_0}e_x + \mathbf{Q_1}T) + (L_1A_1 + L_2 + M)e_x]\,E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] = 0_{m\times 1}\end{aligned}$$

which can be simplified to

$$\begin{aligned}&\left[F\mathbf{R}^2 + G\mathbf{R} + H\right]Y_{t-1} + [[F_1\mathbf{R} + G_1]\,\mathbf{Q_0} + M_1]\,x_t + \\ &[[F_1\mathbf{R} + G_1]\,\mathbf{Q_1} + F_1\mathbf{Q_1}\mathbf{A} + (F_2\mathbf{R} + G_2)\mathbf{Q_1}T + F_2\mathbf{Q_1}T\mathbf{A} \\ &[(F_2\mathbf{R} + G_2)\mathbf{Q_0} + (F_1 + F_2)\mathbf{Q_0}A_1 + (L_1A_1 + L_2)]\,e_x]\,E_t^{(1)}\left[x_t^{(0:\bar{k})}\right] = 0_{m\times 1}\end{aligned}$$

This condition must hold for all realizations of Y_{t-1}, x_t and $E_t^{(1)}\left[x_t^{(0:\bar{k})}\right]$. Therefore, all

coefficients between square brackets must be zero, which leads to

$$
\begin{aligned}
&F\mathbf{R}^2 + G\mathbf{R} + H = 0_{m\times m} \\
&[F_1\mathbf{R} + G_1]\,\mathbf{Q_0} + M_1 = 0_{m\times n} \\
&[F_1\mathbf{R} + G_1]\,\mathbf{Q_1} + F_1\mathbf{Q_1}\mathbf{A} + (F_2\mathbf{R} + G_2)\mathbf{Q_1}T + F_2\mathbf{Q_1}T\mathbf{A} \\
&[(F_2\mathbf{R} + G_2)\mathbf{Q_0} + F\mathbf{Q_0}A_1 + (LA_1 + M_2)]\,e_x = 0_{m\times k}
\end{aligned}
$$

which leads to the same equations from Proposition 2. $\mathbf{R}$ can be solved using Uhlig [2001] method. For a given solution of $\mathbf{R}$, $\mathbf{Q_0}$ and $\mathbf{Q_1}$ can be solved by straightforward vectorization as discussed in the Proposition.

□

D.3 Algorithm

Algorithm. Set the initial values $(\mathbf{A}^{(0)}, \mathbf{B}^{(0)})$, a small tolerance $\epsilon > 0$ and set $i = 1$. Then, follow the steps:

1. Given $\mathbf{A} = \mathbf{A}^{(i-1)}$ and $\mathbf{B}^{(i-1)}$, compute $\bar{\mathbf{K}}$ and $\bar{\mathbf{P}}$ using equations (37) using standard solver for Riccati equations. Set $\bar{\mathbf{K}}^{(i)} = \bar{\mathbf{K}}$ and $\bar{\mathbf{P}}^{(i)} = \bar{\mathbf{P}}$.

2. Given $\mathbf{A}^{(i-1)}$ and $\bar{\mathbf{K}}^{(i)}$, compute the right hand side of equations (36) and solve for $\mathbf{B}$ and $\mathbf{A}$ the equations by matrix inversion. Set $\mathbf{B}^{(i)} = \mathbf{B}$, $\mathbf{A}^{(i)} = \mathbf{A}$.

3. If $\max\left\{||\mathbf{B}^{(i)} - \mathbf{B}^{(i-1)}||, ||\mathbf{A}^{(i)} - \mathbf{A}^{(i-1)}||, ||\bar{\mathbf{P}}^{(i)} - \bar{\mathbf{P}}^{(i-1)}||\right\} < \epsilon$, stop iterating. Otherwise, set $i = i + 1$ and go back to step 1.

Given the solution for $\mathbf{A}$, one can use standard techniques for solving for $(\mathbf{R}, \mathbf{Q_0}, \mathbf{Q_1}, \mathbf{Q})$ using Proposition 2.

E Posterior estimates when shutting down frictions

Table 5 in Section 5.2 of the main paper shows the marginal likelihood of the models shutting down frictions that generate inertia re-estimated for both datasets. In this appendix, we show the details of the posterior estimates of each of those models. The priors are the same as the baseline for all unaffected parameters, with the exception of the exercise where we shut down investment adjustment costs.

If we completely shut down investment adjustment costs, investment would freely adjust to shocks, which is a well-known issue for explaining the investment data. Thus, when shutting down investment adjustment costs, we add capital adjustment costs. In the following,

we briefly show which equations are changed in the model and in the log-linear version. The details of the derivation are in Sections G–I.

CAC model In the baseline model, we use the following law of motion (Equation 10 in the main paper), reproduced here for convenience:

$$K_{h,t} = (1-\delta)K_{h,t-1} + e^{x_t^i}\left(1 - S(I_{h,t}/I_{h,t-1})\right) I_{h,t}, \tag{E.1}$$

where $S(I_t/I_{t-1})$ is the adjustment investment cost function such that $S(\gamma) = S'(\gamma) = 0$, and $S''(\gamma) = s'' > 0$, as in Christiano et al. [2005].

In the model with standard capital adjustment costs in the tradition of Bernanke et al. [1999] and Chari et al. [2002], the law of motion for capital is the following:

$$K_{h,t} = \left(1 - \delta - S_k\left(\frac{I_{h,t}}{K_{h,t-1}}\right)\right) K_{h,t-1} + e^{x_t^i} I_{h,t}. \tag{E.2}$$

Note that $S_k(\cdot)$ is the capital adjustment cost (CAC) function such that $S_k(\gamma - 1 + \delta) = S_k'(\gamma - 1 + \delta) = 0$, and $S_k''(\gamma - 1 + \delta) = s_k'' > 0$. s_k'' is the parameter governing capital adjustment costs. When $\gamma = 1$ (there is no balanced growth for productivity), the specification is the same as in Chari et al. [2002].

In the presence of capital adjustment costs, the first-order conditions for capital and investment are, respectively:

$$\begin{aligned} E_{h,t}[\Phi_{h,t}] &= \beta E_{h,t}\left[\Lambda_{h,t+1}\left(R_{k,t+1}U_{h,t+1} - P_{t+1}a(U_{h,t+1})\right)\right] \\ &+\beta E_{h,t}\left[\Phi_{h,t+1}\left[1 - \delta - S\left(\frac{I_{h,t+1}}{K_{h,t}}\right) + S'\left(\frac{I_{h,t+1}}{K_{h,t}}\right)\frac{I_{h,t+1}}{K_{h,t}}\right]\right] \end{aligned} \tag{E.3}$$

$$E_{h,t}[\Phi_{h,t}e^{x_t^i}] = E_{h,t}[\Lambda_{h,t}P_t]\left(1 + S_k'\left(\frac{I_{h,t}}{K_{h,t-1}}\right)\right). \tag{E.4}$$

In this version, the log-linearized investment equation becomes:

$$i_t = k_{t-1} + \left(\frac{1}{s_k''(\gamma - (1-\delta))}\right) q_t + \bar{E}_t[\hat{x}_t^i] \tag{E.5}$$

and the log-linearized version of equation of the capital pricing equation becomes:

$$\begin{aligned} q_t &= \left(1 - \frac{(1-\delta)\beta}{\gamma}\right)\bar{E}_t\left[r_{t+1}^k\right] + \\ &\left(\frac{(1-\delta)\beta}{\gamma}\right)\int_0^1 E_{h,t}\left[q_{h,t+1}\right]dh + \frac{\beta}{\gamma}s_k''\left(1 - \frac{1-\delta}{\gamma}\right)^2 \bar{E}_t\left[i_{t+1} - k_t\right] - \bar{E}_t[r_t - \pi_{t+1}] \end{aligned} \tag{E.6}$$

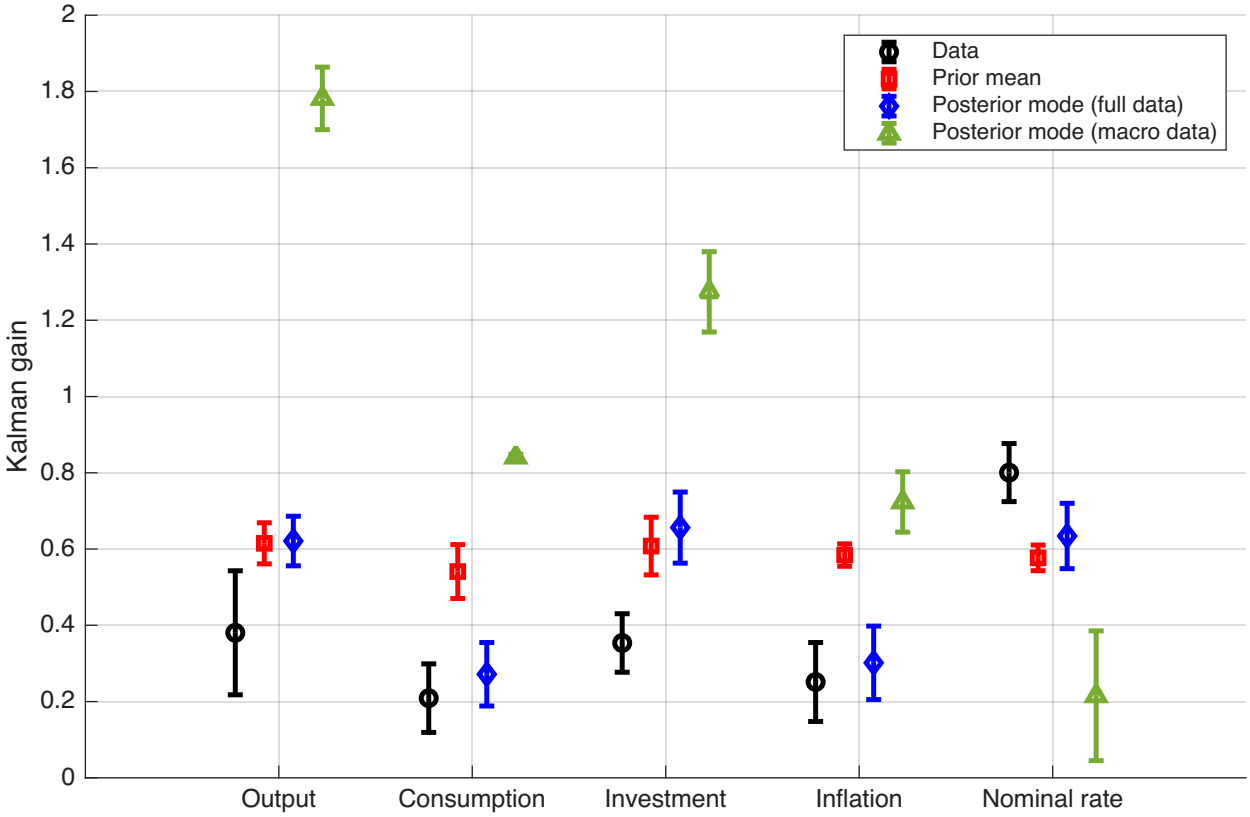
where $q_t = \int_0^1 [q_{h,t}]\, dh$ is the average expected shadow price of capital.

For estimating the CAC model, we need to define a prior distribution for s_k''. We follow the same approach used to choose the prior for IAC. We assume a Gamma distribution with mean 4, which is the distribution and mean from the IAC model for s''. This leads to a standard deviation for investment growth rate that is at par with the baseline model. We decide to use a flatter prior due to the lack of medium-scale DSGE models estimated with CAC, leading to a standard deviation of 2.5 (we use 1.5 for the IAC).

Tables E1 and E2 show the posterior estimates when shutting down standard frictions for the complete dataset and the macroeconomic dataset, respectively.

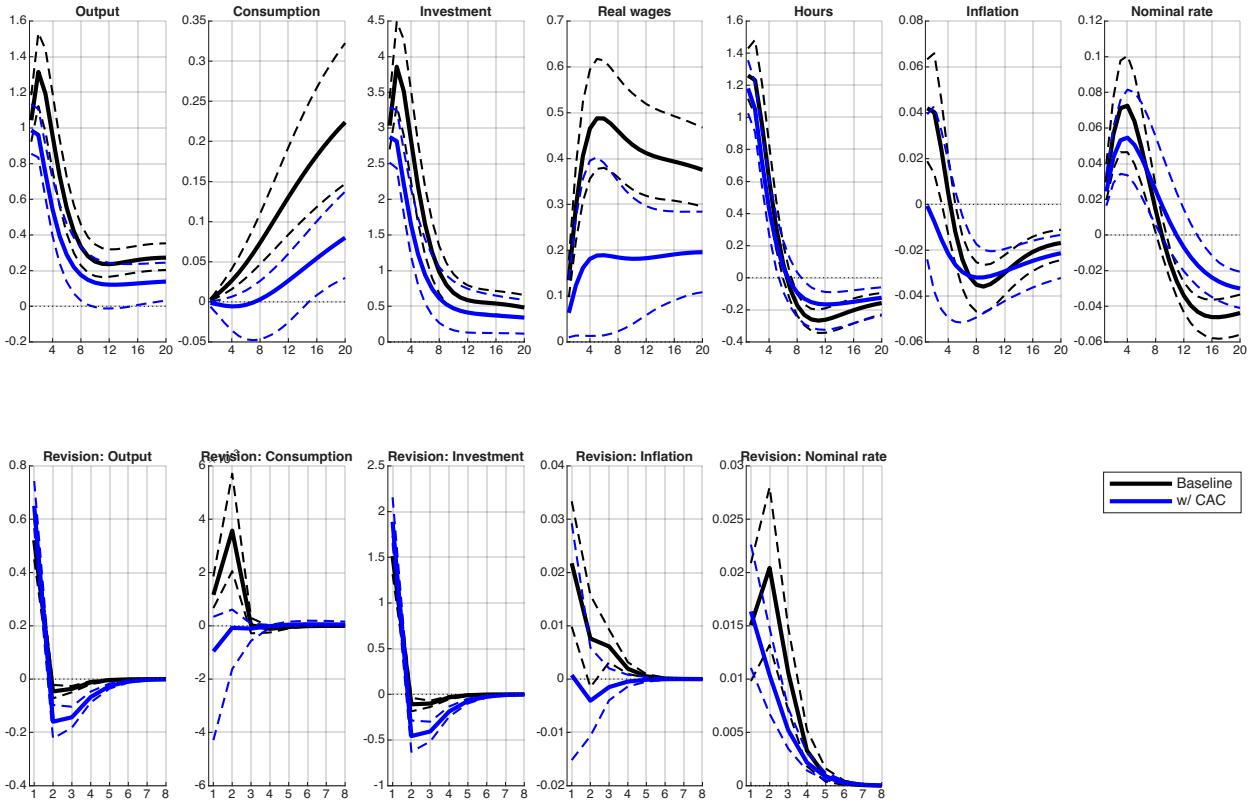
In the main text (Section 5.4 of the main paper), we discussed that the model in which capital adjustment costs replace investment adjustment costs has a fit to the data that is comparable to the baseline model, according to the marginal likelihood criterion. Below, we show that the model with capital adjustment costs fits the empirical Kalman gains closely to the baseline model (Figure E1) and that it also generates similar impulse responses (Figures E2 and E3).

Figure E1: Empirical Kalman gains from DI model with capital adjustment costs



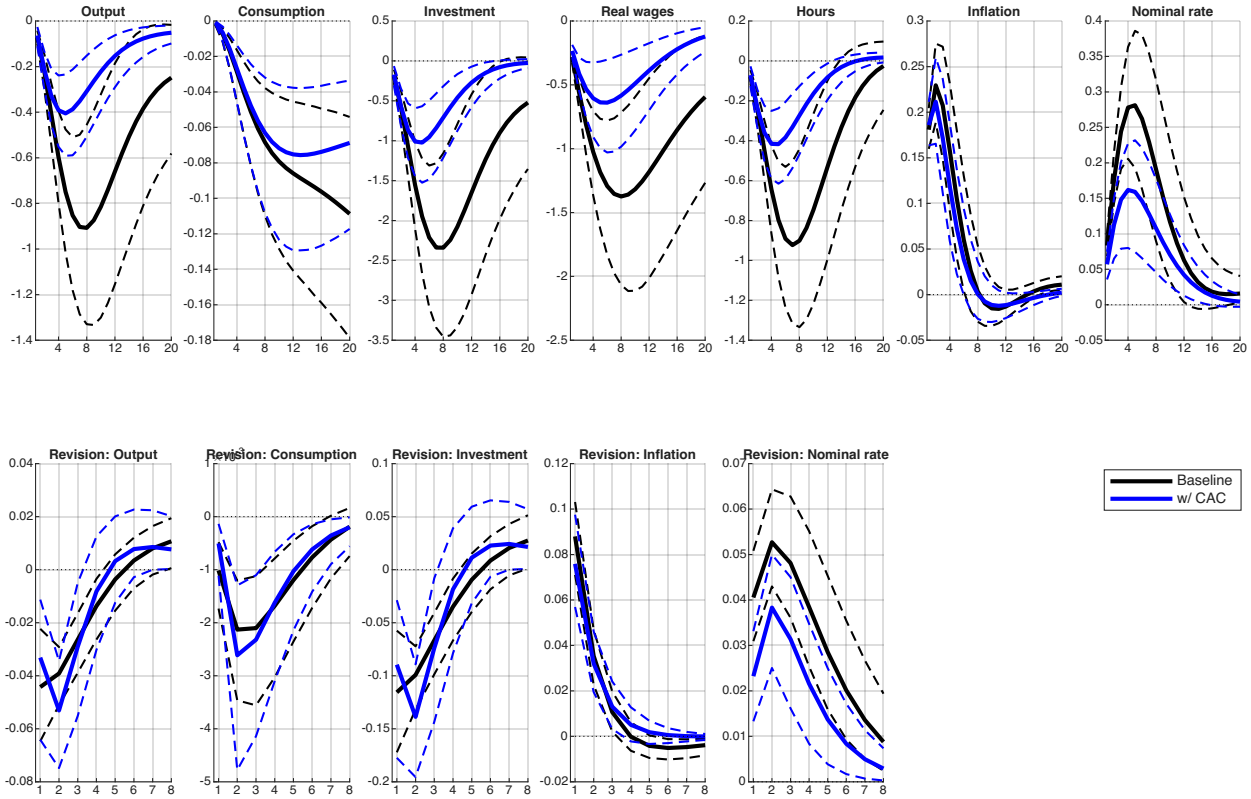
Note: Output, consumption and investment refer to the Kalman gains associated with the expectations about the *growth rate* of the corresponding variables. Inflation and interest rates refer to the Kalman gains associated with the expectations about the *level* of the corresponding variables. The black points correspond to the estimated eKGs from the expectations data. The red, blue and green symbols correspond to the simulated eKGs from the baseline model evaluated at the prior mean, posterior mode using the complete dataset and the posterior mode using macroeconomic data only, respectively. Bars correspond to 90% confidence intervals. Sample: 1981Q4-2007Q4.

Figure E2: Impulse responses to an investment-specific technological shock



Notes: The solid dark line is the impulse response from the baseline model. The solid blue line is the impulse response from the model with capital adjustment costs. Dashed lines are 90% posterior credible sets.

Figure E3: Impulse responses to price mark-up (cost-push) shock



Notes: The solid dark line is the impulse response from the baseline model. The solid blue line is the impulse response from the model with capital adjustment costs. Dashed lines are 90% posterior credible sets.

Table E1: Posterior distributions when shutting down key standard frictions: Complete dataset

	Baseline			No standard frictions			No habits			No inv. adjust. costs			No price indexation			No wage indexation		
Parameters	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%
Endogenous propagation parameters																		
χ	2.52	1.93	3.85	0.86	0.54	1.37	0.78	0.47	1.23	0.80	0.52	4.46	2.49	1.95	3.83	2.51	1.88	4.48
φ	0.99	0.98	0.99	-	-	-	-	-	-	0.99	0.97	0.995	0.99	0.97	0.99	0.99	0.97	0.99
a''	0.28	0.17	0.52	0.43	0.30	0.67	0.19	0.13	0.41	0.27	0.16	0.51	0.28	0.17	0.54	0.31	0.19	0.56
s''	2.68	2.01	3.70	-	-	-	1.64	1.23	2.38	-	-	-	2.59	1.97	3.64	2.62	1.88	3.45
s''_k	-	-	-	6.15	5.07	7.47	-	-	-	26.25	19.61	34.97	-	-	-	-	-	-
ϕ_y	0.06	0.03	0.12	0.58	0.44	0.82	0.33	0.25	0.47	0.15	0.10	0.44	0.06	0.03	0.11	0.08	0.04	0.15
ϕ_π	3.26	2.73	4.22	0.57	0.36	1.01	0.96	0.72	1.59	3.29	2.48	4.15	3.22	2.73	4.18	3.25	2.68	4.23
ϕ_r	0.86	0.82	0.89	0.73	0.59	0.79	0.85	0.77	0.89	0.91	0.87	0.93	0.87	0.82	0.89	0.86	0.82	0.89
ι_p	0.03	0.01	0.09	-	-	-	0.15	0.05	0.35	0.04	0.02	0.16	-	-	-	0.03	0.01	0.12
ι_w	0.62	0.24	0.85	-	-	-	0.19	0.08	0.58	0.47	0.18	0.82	0.56	0.25	0.86	-	-	-
ξ_p	0.79	0.75	0.82	0.95	0.94	0.96	0.95	0.94	0.96	0.88	0.82	0.92	0.79	0.75	0.83	0.81	0.76	0.84
ξ_w	0.69	0.63	0.74	0.77	0.69	0.83	0.78	0.70	0.83	0.80	0.64	0.85	0.69	0.63	0.74	0.71	0.64	0.76
Steady-state parameters																		
π_{ss}	3.04	2.53	3.55	1.92	1.49	2.50	1.58	1.23	2.13	2.18	1.57	2.77	3.08	2.57	3.58	2.94	2.37	3.46
r_{ss}	6.55	5.65	7.43	5.88	5.23	6.64	5.93	4.94	6.89	6.24	5.54	7.16	6.51	5.68	7.51	6.42	5.62	7.50
γ_{ss}	0.28	0.24	0.33	0.47	0.44	0.50	0.50	0.48	0.52	0.44	0.31	0.46	0.29	0.24	0.33	0.30	0.25	0.35
Structural shocks parameters																		
ρ_c	0.15	0.09	0.23	0.94	0.90	0.96	0.95	0.88	0.97	0.13	0.07	0.20	0.16	0.09	0.23	0.14	0.07	0.22
ρ_i	0.30	0.25	0.36	0.89	0.85	0.91	0.39	0.33	0.46	0.74	0.66	0.82	0.30	0.25	0.37	0.30	0.25	0.36
ρ_g	1.00	0.99	1.00	0.97	0.96	0.98	0.97	0.95	0.98	1.00	0.99	1.00	1.00	0.99	1.00	1.00	0.99	1.00
ρ_p	0.87	0.74	0.91	0.37	0.29	0.47	0.34	0.22	0.46	0.68	0.56	0.78	0.87	0.76	0.91	0.74	0.67	0.89
ρ_w	0.18	0.09	0.28	0.24	0.14	0.36	0.28	0.17	0.43	0.29	0.17	0.41	0.17	0.09	0.28	0.24	0.14	0.33
ρ_r	0.48	0.35	0.60	0.90	0.86	0.93	0.72	0.53	0.82	0.39	0.27	0.51	0.48	0.36	0.60	0.49	0.36	0.60
ρ_a	0.93	0.89	0.99	0.99	0.99	1.00	1.00	0.99	1.00	0.99	0.97	1.00	0.93	0.89	0.99	0.99	0.92	1.00
σ_c	2.08	1.56	3.24	2.47	1.94	3.18	3.06	2.02	4.09	1.64	1.29	2.36	2.08	1.58	3.27	2.02	1.52	2.93
σ_i	4.26	3.67	5.26	9.83	8.43	11.58	4.42	3.82	5.28	5.44	4.66	6.78	4.30	3.71	5.26	4.41	3.78	5.37
σ_g	0.56	0.50	0.65	0.59	0.52	0.68	0.59	0.52	0.70	0.56	0.50	0.65	0.56	0.50	0.65	0.56	0.49	0.65
σ_p	0.32	0.24	0.43	0.51	0.36	1.27	0.43	0.30	0.76	0.28	0.20	0.45	0.29	0.23	0.41	0.35	0.27	0.51
σ_w	0.96	0.74	1.74	0.73	0.51	1.38	0.63	0.46	1.34	0.95	0.64	1.70	0.96	0.75	1.78	0.95	0.71	1.70
σ_r	0.13	0.12	0.16	0.15	0.13	0.17	0.11	0.10	0.13	0.13	0.11	0.15	0.13	0.12	0.16	0.13	0.12	0.15
σ_a	0.48	0.43	0.55	0.42	0.38	0.48	0.46	0.42	0.53	0.42	0.38	0.48	0.49	0.43	0.55	0.48	0.43	0.55
Information friction parameters																		
$\bar{k}_c$	0.24	0.18	0.31	0.63	0.59	0.67	0.56	0.52	0.60	0.27	0.22	0.34	0.24	0.18	0.31	0.25	0.19	0.31
$\bar{k}_i$	0.48	0.44	0.53	0.55	0.52	0.58	0.47	0.44	0.51	0.60	0.57	0.65	0.48	0.44	0.53	0.48	0.44	0.52
$\bar{k}_g$	0.55	0.51	0.59	0.66	0.62	0.70	0.53	0.50	0.57	0.61	0.56	0.66	0.55	0.51	0.58	0.56	0.52	0.60
$\bar{k}_p$	0.31	0.25	0.38	0.18	0.09	0.32	0.26	0.17	0.37	0.31	0.23	0.41	0.31	0.25	0.38	0.31	0.24	0.37
$\bar{k}_w$	0.41	0.26	0.54	0.43	0.28	0.68	0.47	0.29	0.74	0.39	0.24	0.56	0.38	0.25	0.53	0.39	0.25	0.54
$\bar{k}_r$	0.69	0.56	0.79	0.43	0.34	0.51	0.39	0.30	0.47	0.72	0.61	0.80	0.70	0.55	0.79	0.64	0.51	0.76
$\bar{k}_a$	0.31	0.07	0.41	0.05	0.03	0.07	0.04	0.03	0.06	0.05	0.03	0.08	0.31	0.08	0.40	0.07	0.05	0.33
Measurement errors standard deviation																		
σ^{me}_{dy}	0.29	0.26	0.33	0.33	0.30	0.37	0.32	0.29	0.36	0.31	0.28	0.35	0.29	0.26	0.33	0.28	0.25	0.32
σ^{me}_{dc}	0.21	0.19	0.23	0.32	0.29	0.37	0.37	0.33	0.41	0.21	0.19	0.23	0.20	0.19	0.23	0.21	0.18	0.23
σ^{me}_{di}	1.22	1.11	1.39	1.36	1.23	1.54	1.23	1.12	1.40	1.38	1.24	1.57	1.23	1.11	1.39	1.22	1.10	1.37
σ^{me}_{π}	0.54	0.48	0.61	0.46	0.42	0.53	0.45	0.40	0.51	0.50	0.44	0.57	0.53	0.48	0.61	0.52	0.47	0.60
σ^{me}_{r}	0.33	0.28	0.39	0.41	0.37	0.48	0.40	0.35	0.47	0.29	0.25	0.34	0.33	0.28	0.39	0.34	0.30	0.41
Marginal Likelihood	- 1,392.60	-	-	- 1,467.20			- 1,473.00			- 1,375.80			- 1,361.50			- 1,376.60		
Number of MH Draws	400k			400k			400k			800k			400k			400k		
MH Burn-in	50%			50%			50%			50%			50%			50%		

Note: $\mathcal{G}$, $\mathcal{IG}$ and $\mathcal{B}$ stand for: Gamma, Inverse Gamma and Beta distributions, respectively.

Table E2: Posterior distributions when shutting down key standard frictions: Macroeconomic dataset

Parameters	Baseline			No standard frictions			No habits			No inv. adjust. costs			No price indexation			No wage indexation		
	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%
Endogenous propagation parameters																		
χ	2.13	1.44	3.53	0.88	0.71	1.55	2.06	1.56	3.22	3.07	1.63	4.83	2.09	1.42	3.50	1.84	1.29	3.39
φ	0.97	0.62	0.98	-	-	-	-	-	-	0.95	0.88	0.97	0.98	0.59	0.99	0.97	0.56	0.98
a''	0.37	0.24	0.61	0.39	0.24	0.65	0.24	0.14	0.49	0.03	0.03	0.04	0.35	0.23	0.60	0.34	0.23	0.60
s''	2.07	1.08	3.65				1.82	1.14	3.16	-	-	-	2.05	1.12	3.73	1.90	1.03	3.51
s''_k	-	-	-	9.44	5.14	14.80	-	-	-	10.18	7.25	14.50	-	-	-	-	-	-
ϕ_y	0.07	0.03	0.14	0.02	0.01	0.05	0.10	0.07	0.17	0.24	0.19	0.36	0.07	0.03	0.14	0.08	0.04	0.15
ϕ_π	2.80	2.23	3.71	1.20	1.15	1.55	2.05	1.73	2.81	1.48	1.26	1.84	2.82	2.27	3.82	2.72	2.15	3.67
ϕ_r	0.87	0.81	0.90	0.60	0.51	0.69	0.79	0.70	0.84	0.70	0.60	0.77	0.86	0.80	0.90	0.86	0.80	0.90
ι_p	0.10	0.04	0.31	-	-	-	0.11	0.04	0.30	0.09	0.04	0.34	-	-	-	0.11	0.04	0.31
ι_w	0.48	0.18	0.80	-	-	-	0.30	0.12	0.74	0.17	0.06	0.51	0.48	0.18	0.81	-	-	-
ξ_p	0.80	0.75	0.83	0.66	0.62	0.71	0.75	0.70	0.79	0.80	0.74	0.85	0.80	0.76	0.84	0.79	0.75	0.83
ξ_w	0.77	0.72	0.81	0.57	0.50	0.65	0.61	0.57	0.69	0.69	0.61	0.78	0.77	0.73	0.81	0.77	0.73	0.81
Steady-state parameters																		
π_{ss}	2.23	1.54	2.92	1.54	1.15	2.17	1.82	1.36	2.69	2.57	1.92	3.43	2.23	1.55	2.95	2.10	1.51	2.87
r_{ss}	6.52	5.68	7.63	7.41	6.46	8.59	6.58	5.75	8.03	5.00	4.36	5.82	6.52	5.69	7.67	6.43	5.69	7.63
γ_{ss}	0.32	0.26	0.37	0.23	0.17	0.33	0.45	0.37	0.49	0.41	0.37	0.45	0.31	0.26	0.37	0.32	0.26	0.37
Structural shocks parameters																		
ρ_c	0.23	0.13	0.85	0.88	0.77	0.98	0.98	0.96	0.99	0.19	0.09	0.30	0.21	0.12	0.87	0.23	0.14	0.88
ρ_i	0.60	0.44	0.88	0.96	0.91	0.98	0.37	0.27	0.48	0.90	0.83	0.96	0.58	0.43	0.85	0.63	0.45	0.92
ρ_g	0.99	0.98	1.00	0.99	0.97	0.997	0.98	0.97	0.99	0.99	0.97	1.00	0.99	0.98	0.997	0.99	0.98	0.997
ρ_p	0.92	0.87	0.95	0.94	0.90	0.97	0.94	0.89	0.97	0.87	0.77	0.93	0.92	0.87	0.95	0.91	0.86	0.95
ρ_w	0.17	0.08	0.28	0.13	0.06	0.22	0.10	0.05	0.21	0.12	0.06	0.22	0.15	0.07	0.27	0.16	0.08	0.28
ρ_r	0.67	0.48	0.86	0.99	0.89	0.99	0.89	0.49	0.95	0.90	0.83	0.96	0.68	0.49	0.88	0.68	0.49	0.86
ρ_a	0.92	0.86	0.97	0.99	0.94	0.997	0.998	0.99	0.999	0.97	0.94	0.99	0.92	0.86	0.97	0.92	0.86	0.98
σ_c	0.35	0.30	0.80	0.71	0.57	1.97	2.45	1.90	6.32	0.35	0.30	0.58	0.35	0.30	0.84	0.38	0.30	0.67
σ_i	1.29	1.05	1.75	6.52	5.22	9.54	1.79	1.44	2.39	2.51	2.00	3.86	1.31	1.05	1.78	1.25	1.02	1.77
σ_g	0.54	0.45	0.63	0.44	0.39	0.53	0.57	0.46	0.66	0.84	0.73	0.99	0.54	0.45	0.63	0.53	0.45	0.62
σ_p	0.10	0.08	0.15	0.32	0.21	0.44	0.12	0.09	0.21	0.07	0.06	0.20	0.09	0.07	0.15	0.10	0.08	0.16
σ_w	0.69	0.46	1.64	1.45	0.99	3.36	0.85	0.57	1.72	1.51	0.98	4.62	0.73	0.51	1.96	0.73	0.48	1.76
σ_r	0.13	0.11	0.14	0.14	0.12	0.16	0.13	0.11	0.15	0.13	0.12	0.15	0.13	0.11	0.15	0.12	0.11	0.14
σ_a	0.41	0.37	0.47	0.42	0.37	0.48	0.43	0.38	0.49	0.41	0.37	0.47	0.41	0.37	0.47	0.41	0.37	0.47
Information friction parameters																		
$\bar{k}_c$	0.94	0.53	0.97	0.91	0.74	0.97	0.72	0.57	0.88	0.85	0.66	0.95	0.92	0.52	0.97	0.94	0.64	0.97
$\bar{k}_i$	0.95	0.82	0.98	0.83	0.44	0.92	0.88	0.73	0.95	0.78	0.49	0.90	0.94	0.82	0.98	0.95	0.83	0.98
$\bar{k}_g$	0.61	0.41	0.80	0.39	0.32	0.52	0.52	0.43	0.69	0.98	0.95	0.99	0.60	0.42	0.81	0.58	0.41	0.80
$\bar{k}_p$	0.88	0.70	0.96	0.49	0.41	0.60	0.83	0.65	0.94	0.80	0.48	0.92	0.84	0.65	0.94	0.88	0.70	0.96
$\bar{k}_w$	0.48	0.25	0.77	0.30	0.16	0.47	0.48	0.26	0.71	0.21	0.09	0.41	0.45	0.21	0.72	0.43	0.23	0.74
$\bar{k}_r$	0.19	0.08	0.60	0.02	0.02	0.11	0.10	0.04	0.25	0.08	0.04	0.17	0.19	0.09	0.63	0.17	0.07	0.57
$\bar{k}_a$	0.67	0.28	0.90	0.34	0.07	0.58	0.03	0.02	0.05	0.02	0.01	0.05	0.70	0.32	0.90	0.73	0.20	0.91
Marginal Likelihood	-950.68			-962.35			-970.40			-928.06			-940.61			-952.35		
Number of MH Draws	400k			1.000k			400k			800k			800k			400k		
MH Burn-in	50%			75%			50%			50%			25%			50%		

Note: $\mathcal{G}$, $\mathcal{IG}$ and $\mathcal{B}$ stand for: Gamma, Inverse Gamma and Beta distributions, respectively.

Table E3: Posterior distributions when shutting down one of each information friction: Complete dataset

Full information for:	Preference Shock			Investment Shock			Gov. expenditure shock			Price markup shock			Wage markup shock			Monetary shock			TFP shock		
Parameters	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%	Mode	5%	95%
Endogenous propagation parameters																					
χ	3.51	2.50	5.14	2.94	2.08	4.71	3.90	2.86	5.81	3.18	2.31	4.79	2.55	2.01	4.27	2.52	1.96	3.46	2.61	1.85	3.76
φ	0.98	0.96	0.98	1.00	0.99	1.00	0.99	0.98	1.00	0.99	0.98	0.99	0.99	0.97	0.99	0.99	0.98	0.99	0.99	0.98	1.00
a''	0.26	0.17	0.51	0.37	0.24	0.64	0.37	0.25	0.64	0.33	0.20	0.58	0.29	0.19	0.55	0.27	0.18	0.54	0.28	0.18	0.53
s''	2.25	1.66	3.12	6.61	5.22	8.66	3.89	2.94	5.21	3.09	2.33	3.88	2.81	2.17	3.85	3.18	2.48	4.08	2.62	1.98	3.54
ϕ_y	0.05	0.03	0.11	0.14	0.06	0.21	0.09	0.04	0.17	0.15	0.10	0.24	0.06	0.03	0.12	0.05	0.03	0.10	0.05	0.03	0.11
ϕ_π	3.50	2.91	4.44	3.47	2.77	4.33	3.34	2.79	4.38	3.27	2.74	4.14	3.17	2.67	4.07	3.29	2.74	4.13	3.25	2.74	4.16
ϕ_r	0.87	0.84	0.90	0.89	0.85	0.91	0.87	0.83	0.90	0.86	0.80	0.89	0.86	0.82	0.89	0.87	0.83	0.90	0.86	0.81	0.89
ι_p	0.03	0.01	0.10	0.04	0.01	0.12	0.04	0.01	0.14	0.02	0.01	0.08	0.03	0.01	0.10	0.03	0.01	0.10	0.03	0.01	0.10
ι_w	0.71	0.36	0.90	0.46	0.18	0.80	0.38	0.15	0.77	0.49	0.19	0.81	0.48	0.17	0.82	0.60	0.26	0.87	0.61	0.25	0.86
ξ_p	0.80	0.76	0.83	0.84	0.80	0.87	0.83	0.78	0.86	0.78	0.75	0.81	0.85	0.82	0.88	0.76	0.73	0.80	0.85	0.82	0.88
ξ_w	0.77	0.74	0.80	0.75	0.69	0.79	0.72	0.67	0.77	0.65	0.60	0.71	0.67	0.60	0.73	0.67	0.62	0.72	0.70	0.66	0.76
Steady-state parameters																					
π_{ss}	3.04	2.58	3.50	2.33	1.71	3.39	3.04	2.46	3.64	2.40	1.83	3.01	3.03	2.48	3.52	3.17	2.74	3.64	3.10	2.57	3.55
r_{ss}	6.28	5.51	7.24	6.84	5.87	8.03	6.97	6.06	8.17	6.31	5.50	7.29	6.53	5.61	7.48	6.48	5.72	7.52	6.35	5.56	7.40
γ_{ss}	0.29	0.25	0.33	0.36	0.31	0.39	0.32	0.27	0.37	0.36	0.31	0.39	0.29	0.24	0.34	0.28	0.23	0.32	0.30	0.26	0.36
Structural shocks parameters																					
ρ_c	0.26	0.16	0.35	0.14	0.08	0.21	0.14	0.08	0.22	0.11	0.05	0.17	0.15	0.08	0.22	0.15	0.09	0.22	0.15	0.09	0.22
ρ_i	0.37	0.31	0.44	0.34	0.28	0.40	0.31	0.24	0.38	0.31	0.24	0.37	0.31	0.25	0.37	0.30	0.24	0.36	0.35	0.28	0.41
ρ_g	0.996	0.99	0.998	0.999	0.98	1.000	0.97	0.96	0.98	0.997	0.99	0.999	0.995	0.99	0.998	0.995	0.99	0.998	0.996	0.99	0.998
ρ_p	0.87	0.80	0.90	0.75	0.68	0.91	0.76	0.68	0.88	0.91	0.86	0.94	0.82	0.67	0.86	0.89	0.84	0.93	0.84	0.78	0.88
ρ_w	0.16	0.08	0.25	0.18	0.09	0.28	0.21	0.10	0.31	0.25	0.15	0.36	0.19	0.07	0.32	0.15	0.08	0.25	0.18	0.10	0.29
ρ_r	0.44	0.31	0.55	0.43	0.31	0.55	0.54	0.40	0.66	0.50	0.37	0.62	0.50	0.37	0.61	0.29	0.20	0.40	0.47	0.36	0.59
ρ_a	0.92	0.87	0.98	0.997	0.92	0.998	0.99	0.98	0.998	0.996	0.99	0.998	0.99	0.91	0.996	0.93	0.89	0.97	0.87	0.82	0.91
σ_c	0.27	0.24	0.31	1.80	1.39	2.73	1.96	1.49	2.85	1.84	1.45	2.71	2.04	1.58	3.36	2.09	1.59	3.24	2.15	1.63	3.41
σ_i	4.04	3.35	4.90	1.27	1.13	1.46	2.86	2.40	3.60	3.63	3.16	4.49	4.33	3.68	5.23	3.99	3.44	4.82	4.10	3.41	4.95
σ_g	0.57	0.51	0.66	0.55	0.46	0.64	0.54	0.48	0.63	0.56	0.49	0.65	0.57	0.49	0.65	0.56	0.50	0.65	0.58	0.51	0.67
σ_p	0.27	0.21	0.37	0.19	0.15	0.30	0.27	0.20	0.40	0.08	0.07	0.10	0.27	0.21	0.39	0.33	0.25	0.45	0.22	0.17	0.30
σ_w	0.83	0.64	1.35	0.94	0.73	2.01	0.93	0.70	1.82	0.78	0.58	1.39	0.40	0.34	0.48	1.05	0.81	1.93	0.91	0.67	1.93
σ_r	0.13	0.12	0.16	0.13	0.12	0.15	0.14	0.12	0.16	0.15	0.13	0.17	0.13	0.12	0.15	0.14	0.12	0.16	0.13	0.12	0.16
σ_a	0.46	0.41	0.52	0.45	0.40	0.52	0.48	0.43	0.56	0.47	0.43	0.55	0.49	0.43	0.56	0.49	0.44	0.56	0.49	0.44	0.57
Information friction parameters																					
$\bar{k}_c$	-	-	-	0.27	0.20	0.33	0.25	0.20	0.32	0.25	0.19	0.31	0.24	0.17	0.30	0.24	0.18	0.30	0.23	0.17	0.30
$\bar{k}_i$	0.49	0.45	0.54	-	-	-	0.62	0.56	0.69	0.53	0.48	0.58	0.48	0.44	0.53	0.51	0.47	0.55	0.50	0.45	0.54
$\bar{k}_g$	0.57	0.53	0.62	0.76	0.70	0.85	-	-	-	0.58	0.55	0.63	0.55	0.51	0.59	0.55	0.51	0.58	0.54	0.50	0.57
$\bar{k}_p$	0.35	0.28	0.43	0.40	0.31	0.51	0.34	0.26	0.43	-	-	-	0.32	0.25	0.40	0.30	0.24	0.38	0.36	0.28	0.46
$\bar{k}_w$	0.45	0.30	0.58	0.35	0.21	0.52	0.40	0.23	0.55	0.49	0.33	0.66	-	-	-	0.36	0.24	0.50	0.38	0.22	0.57
$\bar{k}_r$	0.72	0.61	0.82	0.77	0.64	0.85	0.70	0.58	0.80	0.83	0.71	0.90	0.61	0.49	0.74	-	-	-	0.71	0.60	0.81
$\bar{k}_a$	0.31	0.07	0.44	0.06	0.04	0.43	0.08	0.06	0.25	0.03	0.02	0.04	0.08	0.06	0.38	0.34	0.24	0.44	-	-	-
Measurement errors standard deviation																					
σ_{dy}^{me}	0.33	0.30	0.38	0.48	0.44	0.56	0.44	0.40	0.52	0.33	0.30	0.37	0.29	0.26	0.33	0.30	0.27	0.34	0.32	0.29	0.37
σ_{dc}^{me}	0.44	0.39	0.49	0.21	0.19	0.23	0.21	0.19	0.23	0.21	0.19	0.23	0.21	0.19	0.23	0.21	0.19	0.23	0.21	0.19	0.23
σ_{di}^{me}	1.31	1.17	1.46	2.02	1.84	2.30	1.41	1.27	1.64	1.35	1.23	1.53	1.23	1.11	1.39	1.25	1.14	1.42	1.29	1.17	1.46
σ_π^{me}	0.55	0.49	0.63	0.54	0.47	0.63	0.54	0.48	0.61	0.70	0.63	0.79	0.54	0.49	0.62	0.56	0.50	0.64	0.55	0.49	0.63
σ_r^{me}	0.31	0.26	0.36	0.28	0.25	0.33	0.31	0.27	0.36	0.32	0.28	0.37	0.36	0.31	0.42	0.30	0.27	0.35	0.31	0.27	0.37
Marginal Likelihood	-1392.60			-1496.90			-1457.40			-1451.50			-1429.60			-1392.50			-1377.30		

Note: $\mathcal{G}$, $\mathcal{IG}$ and $\mathcal{B}$ stand for: Gamma, Inverse Gamma and Beta distributions, respectively. All RWMH used 400k draws and 50% burn-in.

F Higher-order truncation

In the baseline estimation reported in Section 4, we truncate the hierarchy of average higher-order expectations at order $\bar{k} = 6$. To assess the sensitivity of the results to this choice, we re-estimate the DI model on the complete dataset using $\bar{k} = 12$.

The marginal likelihood under $\bar{k} = 12$ is essentially unchanged relative to the baseline, indicating that the lower-order truncation already provides an accurate approximation to the equilibrium policy functions. Most posterior estimates are also similar across the two truncation orders, as reported in Table F1.

The main exceptions are the following. First, the investment adjustment cost parameter s'' falls from 2.68 to 1.67. Second, wage indexation ι_w rises from 0.62 to 0.81. Third, the inverse Frisch elasticity χ increases from 2.52 to 4.41. Finally, the Kalman gain for the TFP shock $\bar{k}_a$ falls from 0.31 to 0.04, implying stronger informational frictions about productivity when higher-order expectations are more finely resolved.

G Log-linearized households' problem

G.1 Detrending procedure

Before deriving the log-linearized equations, we discuss the detrending procedure.

All real variables grow along with the productivity trend, so they are detrended as follows: $\hat{Z}_t = \frac{Z_t}{\gamma^t}$, for any real variable Z_t. Nominal variables grow along with the price level P_t, hence they are stationarized using the following procedure: $\hat{Z}_t = \frac{Z_t}{P_t}$.[2]

One exception is that optimal prices and wages are detrended by dividing them by the price level of the last period: $\hat{P}^*_{i,t} = P^*_{i,t}/P_{t-1}$, while $\hat{W}^*_{h,t} = W^*_{h,t}/\gamma^t P_{t-1}$.[3]

Stationary variables are transformed by taking their log-deviation to steady-state value as follows: $z_t = \log(Z_t/\bar{Z})$, for any stationary variable Z_t. Thus, lower case variables denote log-deviation from steady-state of the upper case variables.

[2] One exception is the Lagrange multiplier of the budget constraint, Λ_t, that must be normalized to $\hat{\Lambda}_t = \Lambda_t \gamma^t P_t$, since it reflects the marginal utility of an additional unit of money, which declines as consumption grows at rate γ and the price level rises.

[3] This simplifies the derivation as $E_{h,t}[\hat{W}^*_{h,t}] = \hat{W}^*_{h,t}$ and $E_{i,t}[\hat{P}^*_{i,t}] = \hat{P}^*_{i,t}$ since agents observe P_{t-1}. This would not be true if we detrended them using P_t.

Table F1: Posterior distributions: $\bar{k} = 6$ vs. $\bar{k} = 12$ truncation (complete dataset)

Parameters	Prior			Posterior: $\bar{k} = 6$			Posterior: $\bar{k} = 12$		
	Dist.	Mean	Std. Dev.	Mode	5%	95%	Mode	5%	95%
Endogenous propagation parameters									
χ	$\mathcal{G}$	2.00	0.75	2.52	1.93	3.85	4.41	3.36	6.61
φ	$\mathcal{B}$	0.50	0.20	0.99	0.98	0.99	0.97	0.96	0.98
a''	$\mathcal{G}$	0.20	0.10	0.28	0.17	0.52	0.37	0.24	0.39
s''	$\mathcal{G}$	4.00	1.50	2.68	2.01	3.70	1.67	1.41	2.18
ϕ_y	$\mathcal{G}$	0.20	0.10	0.06	0.03	0.12	0.05	0.03	0.09
ϕ_π	$\mathcal{G}$	1.50	0.50	3.26	2.73	4.22	3.39	2.90	4.30
ϕ_r	$\mathcal{B}$	0.50	0.20	0.86	0.82	0.89	0.82	0.76	0.85
ι_p	$\mathcal{B}$	0.50	0.20	0.03	0.01	0.09	0.04	0.02	0.16
ι_w	$\mathcal{B}$	0.50	0.20	0.62	0.24	0.85	0.81	0.52	0.93
ξ_p	$\mathcal{B}$	0.67	0.05	0.79	0.75	0.82	0.80	0.77	0.83
ξ_w	$\mathcal{B}$	0.75	0.05	0.69	0.63	0.74	0.64	0.59	0.69
Steady-state parameters									
π_{ss}	$\mathcal{G}$	2.62	0.50	3.04	2.53	3.55	1.52	1.10	1.98
r_{ss}	$\mathcal{G}$	5.15	0.50	6.55	5.65	7.43	7.27	6.36	8.17
γ_{ss}	$\mathcal{G}$	0.50	0.10	0.28	0.24	0.33	0.23	0.19	0.28
Structural shocks parameters									
ρ_c	$\mathcal{B}$	0.50	0.20	0.15	0.09	0.23	0.16	0.09	0.23
ρ_i	$\mathcal{B}$	0.50	0.20	0.30	0.25	0.36	0.34	0.28	0.39
ρ_g	$\mathcal{B}$	0.50	0.20	0.996	0.987	0.998	0.990	0.975	0.996
ρ_p	$\mathcal{B}$	0.50	0.20	0.87	0.74	0.91	0.77	0.69	0.82
ρ_w	$\mathcal{B}$	0.50	0.20	0.18	0.09	0.28	0.24	0.14	0.33
ρ_r	$\mathcal{B}$	0.50	0.20	0.48	0.35	0.60	0.55	0.42	0.66
ρ_a	$\mathcal{B}$	0.50	0.20	0.93	0.89	0.99	1.00	1.00	1.00
σ_c	$\mathcal{IG}$	0.50	1.00	2.08	1.56	3.24	1.85	1.47	2.64
σ_i	$\mathcal{IG}$	2.00	1.00	4.26	3.67	5.26	5.14	4.36	6.17
σ_g	$\mathcal{IG}$	0.20	1.00	0.56	0.50	0.65	0.45	0.39	0.52
σ_p	$\mathcal{IG}$	0.20	1.00	0.32	0.24	0.43	0.27	0.21	0.39
σ_w	$\mathcal{IG}$	0.50	1.00	0.96	0.74	1.74	0.96	0.75	1.69
σ_r	$\mathcal{IG}$	0.20	1.00	0.13	0.12	0.16	0.15	0.13	0.18
σ_a	$\mathcal{IG}$	0.20	1.00	0.48	0.43	0.55	0.48	0.43	0.55
Information friction parameters									
$\bar{k}_c$	$\mathcal{B}$	0.50	0.20	0.24	0.18	0.31	0.26	0.21	0.33
$\bar{k}_i$	$\mathcal{B}$	0.50	0.20	0.48	0.44	0.53	0.44	0.40	0.48
$\bar{k}_g$	$\mathcal{B}$	0.50	0.20	0.55	0.51	0.59	0.52	0.49	0.57
$\bar{k}_p$	$\mathcal{B}$	0.50	0.20	0.31	0.25	0.38	0.35	0.29	0.45
$\bar{k}_w$	$\mathcal{B}$	0.50	0.20	0.41	0.26	0.54	0.41	0.28	0.55
$\bar{k}_r$	$\mathcal{B}$	0.50	0.20	0.69	0.56	0.79	0.68	0.56	0.77
$\bar{k}_a$	$\mathcal{B}$	0.50	0.20	0.31	0.07	0.41	0.04	0.04	0.06
Measurement errors standard deviation									
σ_{dy}^{me}	$\mathcal{IG}$	0.10	0.02	0.29	0.26	0.33	0.28	0.25	0.32
σ_{dc}^{me}	$\mathcal{IG}$	0.08	0.02	0.21	0.19	0.23	0.21	0.19	0.23
σ_{di}^{me}	$\mathcal{IG}$	0.50	0.10	1.22	1.11	1.39	1.22	1.09	1.36
σ_{π}^{me}	$\mathcal{IG}$	0.15	0.04	0.54	0.48	0.61	0.53	0.47	0.60
σ_{r}^{me}	$\mathcal{IG}$	0.15	0.04	0.33	0.28	0.39	0.35	0.30	0.40
Marginal Likelihood				−1392.60			−1391.90		

Note: $\mathcal{G}$, $\mathcal{IG}$ and $\mathcal{B}$ stand for: Gamma, Inverse Gamma and Beta distributions, respectively. Both models are estimated on the complete dataset (macroeconomic and expectations data).

G.2 Euler Equation

We first detrend and log-linearize equations (G.1) and (G.2) to get the usual Euler equation for consumption:

$$E_{h,t}[\Lambda_{h,t}P_t] = \frac{E_{h,t}\left[e^{x_t^c}\right]}{C_{h,t} - \varphi C_{h,t-1}} \tag{G.1}$$

$$E_{h,t}[\Lambda_{h,t}] = \beta E_{h,t}\left[\Lambda_{h,t+1}R_t\right]. \tag{G.2}$$

Detrending equation (G.1), we get

$$E_{h,t}\left[\gamma^t \Lambda_{h,t}P_t\right] = \frac{E_{h,t}\left[e^{x_t^c}\right]}{C_{h,t}/\gamma^t - \varphi C_{h,t-1}/(\gamma \cdot \gamma^{t-1})}$$

$$E_{h,t}[\hat{\Lambda}_{h,t}] = \frac{E_{h,t}\left[e^{x_t^c}\right]}{\hat{C}_{h,t} - \varphi/\gamma \hat{C}_{h,t-1}}$$

Log-linearizing the equation above, it follows that

$$\bar{\Lambda}_h(1 + E_{h,t}[\lambda_{h,t}]) = 1 + \frac{E_{h,t}[x_t^c]}{\bar{C}_h[1 - \varphi/\gamma + c_{h,t} - \varphi/\gamma c_{h,t-1}]}$$

Using the steady-state result $\bar{\Lambda}_h = \left[\bar{C}_h\left(1 - \varphi/\gamma\right)\right]^{-1}$ we find

$$(1 + E_{h,t}[\lambda_{h,t}])\left(1 - \frac{\varphi}{\gamma} + c_{h,t} - \frac{\varphi}{\gamma}c_{h,t-1}\right) = \left(1 - \frac{\varphi}{\gamma}\right)(1 + E_{h,t}\left[x_t^c\right])$$

Rearranging terms

$$E_{h,t}[\lambda_{h,t}] = \frac{\varphi/\gamma}{1 - \varphi/\gamma}c_{h,t-1} - \frac{1}{1 - \varphi/\gamma}c_{h,t} + E_{h,t}\left[x_t^c\right] \tag{G.3}$$

The detrended version of (G.2) is

$$E_{h,t}[\gamma^t \Lambda_{h,t}P_t] = \frac{\beta}{\gamma}E_{h,t}\left[\gamma^{t+1}\Lambda_{h,t+1}P_{t+1}\frac{P_t}{P_{t+1}}R_t\right]$$

$$E_{h,t}[\hat{\Lambda}_{h,t}] = \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Lambda}_{h,t+1}\frac{R_t}{\Pi_{t+1}}\right]$$

We now log-linearize the equation above

$$\bar{\Lambda}_h(1 + E_{h,t}[\lambda_{h,t}]) = \frac{\beta}{\gamma}\frac{\bar{R}}{\bar{\Pi}}\bar{\Lambda}_h E_{h,t}\left[1 + \lambda_{h,t+1} + r_t - \pi_{t+1}\right]$$

Using the steady-state result that $\bar{\Pi}/\bar{R} = \beta/\gamma$ and rearranging terms

$$E_{h,t}[\lambda_{h,t}] = E_{h,t}[\lambda_{h,t+1} + r_t - \pi_{t+1}] \tag{G.4}$$

Substituting (G.3) into (G.4) implies

$$\frac{\varphi/\gamma}{1-\varphi/\gamma}c_{h,t-1} - \frac{1}{1-\varphi/\gamma}c_{h,t} + \pi_t - E_{h,t}[\pi_t] + E_{h,t}\left[x_t^c\right] = \pi_t - E_{h,t}[\pi_t] +$$
$$E_{h,t}\left[\frac{\varphi/\gamma}{1-\varphi/\gamma}c_{h,t} - \frac{1}{1-\varphi/\gamma}c_{h,t+1} + \pi_{t+1} - E_{h,t}[\pi_{t+1}] + E_{h,t}\left[x_{t+1}^c\right] + r_t - \pi_{t+1}\right]$$

Rearranging terms

$$c_{h,t} = \frac{\varphi/\gamma}{1+\varphi/\gamma}c_{h,t-1} + \frac{1}{1+\varphi/\gamma}E_{h,t}[c_{h,t+1}] - \frac{1-\varphi/\gamma}{1+\varphi/\gamma}E_{h,t}[r_t - \pi_{t+1} - \left(x_{t+1}^c - x_t^c\right)]$$

Using the result $c_{h,t} = c_t$ for all h and integrating over h, we get the log-linearized Euler equation

$$c_t = \frac{\varphi/\gamma}{1+\varphi/\gamma}c_{t-1} + \frac{1}{1+\varphi/\gamma}\bar{E}_t[c_{t+1}] - \frac{1-\varphi/\gamma}{1+\varphi/\gamma}\bar{E}_t[r_t - \pi_{t+1} - \left(x_{t+1}^c - x_t^c\right)] \tag{G.5}$$

G.3 Equilibrium condition for investment

In the baseline model, we assume that investment decisions are subject to an investment adjustment cost. However, in a later exercise, we consider adjustment costs in capital instead. This section derives the investment equilibrium condition for both cases.

Investment adjustment costs. The first-order condition for investment (G.6) with adjustment costs in investment is the following:

$$\begin{aligned} E_{h,t}[\Lambda_{h,t}P_t] = E_{h,t}\left[\Phi_{h,t}e^{x_t^i}\right]\left(1 - S\left(\frac{I_{h,t}}{I_{h,t-1}}\right) - S'\left(\frac{I_{h,t}}{I_{h,t-1}}\right)\frac{I_{h,t}}{I_{h,t-1}}\right) \\ +\beta E_{h,t}\left[\Phi_{h,t+1}e^{x_{t+1}^i}S'\left(\frac{I_{h,t+1}}{I_{h,t}}\right)\left(\frac{I_{h,t+1}}{I_{h,t}}\right)^2\right] \end{aligned} \tag{G.6}$$

We start by detrending the equation above

$$E_{h,t}[\hat{\Lambda}_{h,t}] = E_{h,t}\left[\hat{\Phi}_{h,t}e^{x_t^i}\right]\left(1 - S\left(\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right) - S'\left(\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right)\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right) + \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Phi}_{h,t+1}e^{x_{t+1}^i}S'\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{I}_{h,t}}\right)\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{I}_{h,t}}\right)^2\right]$$

Using the definition $Q_{h,t} \equiv E_{h,t}[\hat{\Phi}_{h,t}/\hat{\Lambda}_{h,t}]$, it follows that

$$E_{h,t}[\hat{\Lambda}_{h,t}] = Q_{h,t}E_{h,t}[\hat{\Lambda}_{h,t}e^{x_t^i}]\left(1 - S\left(\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right) - S'\left(\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right)\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right) + \frac{\beta}{\gamma}E_{h,t}\left[Q_{h,t+1}\hat{\Lambda}_{h,t+1}e^{x_{t+1}^i}S'\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{I}_{h,t}}\right)\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{I}_{h,t}}\right)^2\right]$$

We now log-linearize the equation above using the following approximations

$$\begin{aligned}
\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}} &\approx 1 + i_{h,t} - i_{h,t-1} \\
S\left(\gamma(1 + i_{h,t} - i_{h,t-1})\right) &\approx S(\gamma) + \gamma S'(\gamma)\left(i_{h,t} - i_{h,t-1}\right) = 0 \\
S'\left(\gamma(1 + i_{h,t+1} - i_{h,t})\right) &\approx S'(\gamma) + \gamma S''(\gamma)(i_{h,t+1} - i_{h,t}) = \gamma s''(i_{h,t+1} - i_{h,t})
\end{aligned}$$

The log-linearized version of the equilibrium condition for investment is

$$1 + E_{h,t}[\lambda_{h,t}] = \bar{Q}_h\left(1 + q_{h,t} + E_{h,t}[\lambda_{h,t}] + E_{h,t}\left[x_t^i\right]\right)\left(1 - \gamma^2 s''(i_{h,t} - i_{h,t-1})(1 + i_{h,t} - i_{h,t-1})\right) + \bar{Q}_h\beta\gamma^2 E_{h,t}\left[(1 + \lambda_{h,t+1})(1 + x_{t+1}^i)s''(i_{h,t+1} - i_{h,t})(1 + i_{h,t} - i_{h,t-1})^2\right]$$

Using the steady-state result $\bar{Q}_h = 1$ and dropping second-order terms off, it follows that

$$E_{h,t}[\lambda_{h,t}] = q_{h,t} + E_{h,t}[\lambda_{h,t}] + E_{h,t}[x_t^i] - \gamma^2 s''(i_{h,t} - i_{h,t-1}) + \beta\gamma^2 s''(E_{h,t}[i_{h,t+1}] - i_{h,t})$$

Rearranging terms, we get

$$i_{h,t} = \left(\frac{1}{1+\beta}\right)i_{h,t-1} + \left(\frac{\beta}{1+\beta}\right)E_{h,t}[i_{h,t+1}] + \left(\frac{1}{s''(1+\beta)\gamma^2}\right)\left(q_{h,t} + E_{h,t}\left[x_t^i\right]\right)$$

Using the fact that $i_{h,t} = i_t$ for all h and t and integrating over h, we get

$$i_t = \left(\frac{1}{1+\beta}\right)i_{t-1} + \left(\frac{\beta}{1+\beta}\right)\bar{E}_t[i_{t+1}] + \left(\frac{1}{s''(1+\beta)\gamma^2}\right)\left(q_t + \bar{E}_t\left[x_t^i\right]\right).$$

Finally, we normalize the shock as follows

$$i_t = \left(\frac{1}{1+\beta}\right) i_{t-1} + \left(\frac{\beta}{1+\beta}\right) \bar{E}_t[i_{t+1}] + \left(\frac{1}{s''(1+\beta)\gamma^2}\right) q_t + \bar{E}_t\left[\hat{x}_t^i\right], \tag{G.7}$$

where $\hat{x}_t^i \equiv x_t^i/(s''(1+\beta)\gamma^2)$.

Capital adjustment cost. In the case with capital adjustment costs, the first-order condition for investment is

$$E_{h,t}[\Phi_{h,t} e^{x_t^i}] = E_{h,t}[\Lambda_{h,t} P_t]\left(1 + S'\left(\frac{I_{h,t}}{K_{h,t-1}}\right)\right).$$

Detrending and using the definition $Q_{h,t} \equiv E_{h,t}[\hat{\Phi}_{h,t}/\hat{\Lambda}_{h,t}]$ yields

$$Q_{h,t} E_{h,t}[\hat{\Lambda}_{h,t} e^{x_t^i}] = E_{h,t}[\hat{\Lambda}_{h,t}]\left(1 + S'\left(\gamma \frac{\hat{I}_{h,t}}{\hat{K}_{h,t-1}}\right)\right).$$

We log-linearize the equation above using the following first-order approximation around the steady state

$$S'\left(\gamma \frac{\hat{I}_{h,t}}{\hat{K}_{h,t-1}}\right) \approx S'\left(\gamma \frac{\bar{I}}{\bar{K}}\right) + s''\left(\gamma \frac{\hat{I}_{h,t}}{\hat{K}_{h,t-1}} - \gamma \frac{\bar{I}}{\bar{K}}\right) \approx s'' \cdot \gamma \frac{\bar{I}}{\bar{K}}(i_{h,t} - k_{h,t-1})$$

Using the steady-state result $\bar{I}_h/\bar{K}_h = 1 - (1-\delta)/\gamma$, it follows that

$$q_{h,t} + E_{h,t}[x_t^i] = s''\left(\gamma - (1-\delta)\right)(i_{h,t} - k_{h,t-1})$$

Rewriting in terms of investment yields

$$i_t = k_{h,t-1} + \left(\frac{1}{s''(\gamma - (1-\delta))}\right) q_{h,t} + E_{h,t}[\hat{x}_t^i]$$

after we use the normalized shock $\hat{x}_t^i \equiv \frac{x_t^i}{s''(\gamma-(1-\delta))}$. Finally, we integrate over h to get

$$i_t = k_{t-1} + \left(\frac{1}{s''(\gamma - (1-\delta))}\right) q_t + \bar{E}_t[\hat{x}_t^i] \tag{G.8}$$

G.4 Asset pricing equation for capital stock

In the baseline model, we assume that investment decisions are subject to an investment adjustment cost. However, in a later exercise, we consider adjustment costs in capital instead.

This section derives the investment equilibrium condition for both cases.

Investment adjustment costs. We use the households' first-order conditions for capital (G.9) below to get an asset pricing equation for capital:

$$E_{h,t}[\Phi_{h,t}] = \beta E_{h,t}\left[\Lambda_{h,t+1}\left(R^k_{t+1}U_{h,t+1} - P_{t+1}a(U_{h,t+1})\right)\right] + (1-\delta)\beta E_{h,t}\left[\Phi_{h,t+1}\right] \tag{G.9}$$

We detrend all variables in (G.9),

$$E_{h,t}\left[\hat{\Phi}_{h,t}\right] = \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Lambda}_{h,t+1}\left(\hat{R}^k_{t+1}U_{h,t+1} - a(U_{h,t+1})\right)\right] + (1-\delta)\frac{\beta}{\gamma}E_{h,t}\left[\hat{\Phi}_{h,t+1}\right]$$

Using the definition $Q_{h,t} \equiv E_{h,t}[\Phi_{h,t}/\Lambda_{h,t}P_t]$, it follows that

$$Q_{h,t}E_{h,t}\left[\hat{\Lambda}_{h,t}\right] = \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Lambda}_{h,t+1}\left(\hat{R}^k_{t+1}U_{h,t+1} - a(U_{h,t+1})\right)\right] + (1-\delta)\frac{\beta}{\gamma}E_{h,t}\left[Q_{h,t+1}\hat{\Lambda}_{h,t+1}\right].$$

Note that $Q_{h,t}$ can be taken outside the expectation operator $E_{h,t}[\cdot]$ by the Law of Iterated Expectations.

We start log-linearizing using the following approximation

$$a(U_{h,t+1}) \approx a(\bar{U}_h) + a'(\bar{U}_h)(U_{h,t+1} - \bar{U}_h) = \bar{R}^k(U_{h,t+1} - 1) \approx \bar{R}^k u_{h,t+1}$$

and use the following steady-state results to simplify the equation: $\bar{Q} = 1$, $\bar{U} = 1$, $\bar{R}^k = \frac{\gamma}{\beta} - (1-\delta)$. It follows that

$$\begin{aligned} 1 + q_{h,t} + E_{h,t}[\lambda_{h,t}] = &\left(1 - \frac{(1-\delta)\beta}{\gamma}\right)\left(1 + E_{h,t}\left[\lambda_{h,t+1} + r^k_{t+1}\right]\right) + \\ &\left(\frac{(1-\delta)\beta}{\gamma}\right)\left(1 + E_{h,t}\left[q_{h,t+1} + \lambda_{h,t+1}\right]\right) \end{aligned}$$

Simplifying further yields

$$q_{h,t} = \left(1 - \frac{(1-\delta)\beta}{\gamma}\right)E_{h,t}\left[r^k_{t+1}\right] + \left(\frac{(1-\delta)\beta}{\gamma}\right)E_{h,t}[q_{h,t+1}] + E_{h,t}[\lambda_{h,t+1}] - \lambda_{h,t}$$

Plugging the log-linearized version of the first-order condition for bonds (G.4) into the expression above, it follows that

$$q_{h,t} = \left(\frac{\beta(1-\delta)}{\gamma}\right)E_{h,t}[q_{h,t+1}] + \left(1 - \frac{\beta(1-\delta)}{\gamma}\right)E_{h,t}\left[r^k_{t+1}\right] - E_{h,t}[r_t - \pi_{t+1}]$$

Integrating over h, we finally get

$$q_t = \left(\frac{\beta(1-\delta)}{\gamma}\right)\int_0^1 E_{h,t}[q_{h,t+1}]dh + \left(1 - \frac{\beta(1-\delta)}{\gamma}\right)\bar{E}_t\left[r^k_{t+1}\right] - \bar{E}_t[r_t - \pi_{t+1}] \tag{G.10}$$

Capital adjustment cost. In the case with capital adjustment costs, the first-order condition for capital is

$$\begin{aligned} E_{h,t}[\Phi_{h,t}] &= \beta E_{h,t}\left[\Lambda_{h,t+1}\left(R_{k,t+1}U_{h,t+1} - P_{t+1}a(U_{h,t+1})\right)\right] \\ &+\beta E_{h,t}\left[\Phi_{h,t+1}\left[1-\delta - S\left(\frac{I_{h,t+1}}{K_{h,t}}\right) + S'\left(\frac{I_{h,t+1}}{K_{h,t}}\right)\frac{I_{h,t+1}}{K_{h,t}}\right]\right] \end{aligned}$$

Detrending the equation above yields

$$\begin{aligned} E_{h,t}[\hat{\Phi}_{h,t}] &= \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Lambda}_{h,t+1}\left(\hat{R}_{k,t+1}U_{h,t+1} - a(U_{h,t+1})\right)\right] \\ &+\frac{\beta}{\gamma}E_{h,t}\left[\hat{\Phi}_{h,t+1}\left[1-\delta - S\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right) + S'\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right)\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right]\right] \end{aligned}$$

Using the definition $Q_{h,t} \equiv E_{h,t}[\Phi_{h,t}/\Lambda_{h,t}P_t]$, it follows that

$$\begin{aligned} Q_{h,t}E_{h,t}\left[\hat{\Lambda}_{h,t}\right] &= \frac{\beta}{\gamma}E_{h,t}\left[\hat{\Lambda}_{h,t+1}\left(\hat{R}^k_{t+1}U_{h,t+1} - a(U_{h,t+1})\right)\right] \\ &+\frac{\beta}{\gamma}E_{h,t}\left[Q_{h,t+1}\hat{\Lambda}_{h,t+1}\left[1-\delta - S\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right) + S'\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right)\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right]\right]. \end{aligned}$$

We start log-linearizing the equation above using the following first-order approximations around the steady state:

$$S\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right) \approx 0$$

$$S'\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}}\right) \approx S'\left(\gamma\frac{\bar{I}}{\bar{K}}\right) + s''\left(\gamma\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}} - \gamma\frac{\bar{I}}{\bar{K}}\right) \approx s''\cdot\gamma\frac{\bar{I}}{\bar{K}}(i_{h,t+1} - k_{h,t})$$

$$\frac{\hat{I}_{h,t+1}}{\hat{K}_{h,t}} \approx \frac{\bar{I}}{\bar{K}}(1 + i_{h,t+1} - k_{h,t})$$

We also use the same first-order approximation for utilization capacity and the same steady-state results as before to get

$$q_{h,t} + E_{h,t}[\lambda_{h,t}] = \left(1 - \frac{(1-\delta)\beta}{\gamma}\right)\left(E_{h,t}\left[\lambda_{h,t+1} + r^k_{t+1}\right]\right) +$$
$$\left(\frac{(1-\delta)\beta}{\gamma}\right)\left(E_{h,t}\left[q_{h,t+1} + \lambda_{h,t+1}\right]\right) + \frac{\beta}{\gamma}s''\left(1 - \frac{1-\delta}{\gamma}\right)^2 E_{h,t}\left[i_{h,t+1} - k_{h,t}\right]$$

Using the log-linearized version of the first-order condition for bonds (G.4), it follows that

$$q_{h,t} = \left(1 - \frac{(1-\delta)\beta}{\gamma}\right) E_{h,t}\left[r^k_{t+1}\right] +$$
$$\left(\frac{(1-\delta)\beta}{\gamma}\right) E_{h,t}\left[q_{h,t+1}\right] + \frac{\beta}{\gamma}s''\left(1 - \frac{1-\delta}{\gamma}\right)^2 E_{h,t}\left[i_{h,t+1} - k_{h,t}\right] - E_{h,t}[r_t - \pi_{t+1}]$$

Integrating over h and using the fact that investment and capital decisions are identical across households finally yields

$$q_t = \left(1 - \frac{(1-\delta)\beta}{\gamma}\right) \bar{E}_t\left[r^k_{t+1}\right] +$$
$$\left(\frac{(1-\delta)\beta}{\gamma}\right) \int_0^1 E_{h,t}\left[q_{h,t+1}\right] dh + \frac{\beta}{\gamma}s''\left(1 - \frac{1-\delta}{\gamma}\right)^2 \bar{E}_t\left[i_{t+1} - k_t\right] - \bar{E}_t[r_t - \pi_{t+1}] \tag{G.11}$$

Note that, as $\gamma \approx 1$, $(\bar{I}/\bar{K})^2 \approx \delta^2 \approx 0$.

G.5 Law of motion of capital

In the baseline model, we assume that investment decisions are subject to an investment adjustment cost. However, in a later exercise, we consider adjustment costs in capital instead. This section derives the law of motion of capital for both cases.

Investment adjustment costs. The law of motion of capital when there are adjustment costs in investment is the following

$$K_{h,t} = (1-\delta)K_{h,t-1} + e^{x^i_t}(1 - S(I_{h,t}/I_{h,t-1}))I_{h,t} \tag{G.12}$$

Detrending the equation above yields

$$\hat{K}_{h,t} = \left(\frac{1-\delta}{\gamma}\right)\hat{K}_{h,t-1} + e^{x^i_t}\left(1 - S\left(\gamma\frac{\hat{I}_{h,t}}{\hat{I}_{h,t-1}}\right)\right)\hat{I}_{h,t}$$

We log-linearize the equation

$$\bar{K}_h(1+k_{h,t}) = \left(\frac{1-\delta}{\gamma}\right)\bar{K}_h(1+k_{h,t-1}) + (1+x_t^i)\left(1 - S\left(\gamma(1+i_{h,t}-i_{h,t-1})\right)\right)\bar{I}_h(1+i_{h,t})$$

Using the steady-state result $\bar{I}_h/\bar{K}_h = 1-(1-\delta)/\gamma$

$$1+k_{h,t} = \left(\frac{1-\delta}{\gamma}\right)(1+k_{h,t-1}) + (1+x_t^i)\left(1 - S\left(\gamma(1+i_{h,t}-i_{h,t-1})\right)\right)\left(1-\frac{1-\delta}{\gamma}\right)(1+i_{h,t})$$

$$k_{h,t} = \left(\frac{1-\delta}{\gamma}\right)k_{h,t-1} + \left(1-\frac{1-\delta}{\gamma}\right)\left(i_{h,t}+x_t^i - S\left(\gamma(1+i_{h,t}-i_{h,t-1})\right)\left(1+i_{h,t}+x_t^i\right)\right)$$

We use the following first-order approximation around the steady state

$$S\left(\gamma(1+i_{h,t}-i_{h,t-1})\right) \approx S(\gamma) + \gamma S'(\gamma)\left(i_{h,t}-i_{h,t-1}\right) = 0,$$

which yields

$$k_{h,t} = \left(\frac{1-\delta}{\gamma}\right)k_{h,t-1} + \left(1-\frac{1-\delta}{\gamma}\right)\left(i_{h,t}+x_t^i\right).$$

Using the normalized shock $\hat{x}_t^i \equiv x_t^i/(s''(1+\beta)\gamma^2)$ and integrating over h, it follows that

$$k_t = \left(\frac{1-\delta}{\gamma}\right)k_{t-1} + \left(1-\frac{1-\delta}{\gamma}\right)i_t + \left[\left(1-\frac{1-\delta}{\gamma}\right)s''(1+\beta)\gamma^2\right]\hat{x}_t^i. \tag{G.13}$$

Capital adjustment costs. The law of motion of capital when there are adjustment costs in capital is the following

$$K_{h,t} = \left(1-\delta-S\left(\frac{I_{h,t}}{K_{h,t-1}}\right)\right)K_{h,t-1} + e^{x_t^i}I_{h,t}$$

Detrending the equation above yields

$$\hat{K}_{h,t} = \frac{1}{\gamma}\left(1-\delta-S\left(\gamma\frac{\hat{I}_{h,t}}{\hat{K}_{h,t-1}}\right)\right)\hat{K}_{h,t-1} + e^{x_t^i}\hat{I}_{h,t}$$

We log-linearize the equation above as follows

$$\bar{K}_h(1+k_{h,t}) = \frac{1}{\gamma}\left[1-\delta-S\left(\gamma(1+i_{h,t}-k_{h,t-1})\right)\right]\bar{K}_h(1+k_{h,t-1}) + \bar{I}_h(1+i_{h,t}+x_t^i)$$

Using the steady-state result $\bar{I}_h = (1-(1-\delta)/\gamma)\bar{K}_h$, we get

$$k_{h,t} = \frac{1}{\gamma}\left[1-\delta - S\left(\gamma(1+i_{h,t}-k_{h,t-1})\right)\right]k_{h,t-1} + \left(1-\frac{1-\delta}{\gamma}\right)\left(i_{h,t}+x_t^i\right)$$

We use the following first-order approximation around the steady state

$$S\left(\gamma(1+i_{h,t}-k_{h,t-1})\right) \approx S(\gamma) + \gamma S'(\gamma)\left(i_{h,t}-k_{h,t-1}\right) = 0,$$

which yields

$$k_{h,t} = \left(\frac{1-\delta}{\gamma}\right)k_{h,t-1} + \left(1-\frac{1-\delta}{\gamma}\right)\left(i_{h,t}+x_t^i\right)$$

Using the normalized shock $\hat{x}_t^i \equiv \frac{x_t^i}{s''(\gamma-(1-\delta))}$ and integrating over h, it follows that

$$k_t = \left(\frac{1-\delta}{\gamma}\right)k_{t-1} + \left(1-\frac{1-\delta}{\gamma}\right)i_t + \left[\left(1-\frac{1-\delta}{\gamma}\right)s''\left(\gamma-(1-\delta)\right)\right]\hat{x}_t^i, \tag{G.14}$$

G.6 Capital utilization level

The optimal value of the capital utilization rate solves

$$E_{h,t}[R_t^k] = a'(U_{h,t})E_{h,t}[P_t]. \tag{G.15}$$

Based on equation (G.15), we obtain a log-linearized version of the capital utilization function by first dividing both sides of the equation by P_t

$$a'(U_{h,t})E_{h,t}\left[\frac{P_t}{P_{t-1}}\right] = E_{h,t}\left[\hat{R}_t^k\frac{P_t}{P_{t-1}}\right]$$

We now log-linearize each term and use the following approximation

$$a'(U_{h,t}) \approx a'(\bar{U}_h) + a''(\bar{U}_h)(U_{h,t}-\bar{U}_h) = \bar{R}^k + a''u_{h,t}$$

This leads to

$$\bar{\Pi}(\bar{R}^k + a''u_{h,t} + \bar{R}^k E_{h,t}[\pi_t]) = \bar{R}^k\bar{\Pi}(1+E_{h,t}[r_t^k+\pi_t])$$
$$u_{h,t} = \left(\bar{R}^k/a''\right)E_{h,t}[r_t^k]$$

Integrating over h, we have

$$u_t = (\bar{R}^k/a'')\bar{E}_t[r_t^k] \tag{G.16}$$

G.7 Utilized capital level

We log-linearize the capital utilization function $K_{h,t}^u = U_{h,t}K_{h,t-1}$ as follows

$$\begin{aligned} K_{h,t}^u/\gamma^t &= U_{h,t}K_{h,t-1}/(\gamma \cdot \gamma^t) \\ \hat{K}_{h,t}^u &= U_{h,t}\hat{K}_{h,t-1}/\gamma \\ k_{h,t}^u &= u_{h,t} + k_{h,t-1} \end{aligned}$$

Integrating over $h \in [0,1]$, we get

$$k_t^u = u_t + k_{t-1} \tag{G.17}$$

G.8 Marginal cost of labor

We detrend the marginal rate of substitution between consumption and leisure

$$MRS_{h,t} = L_{h,t}^{\chi}\left(C_{h,t} - \varphi C_{h,t-1}\right)$$

as follows

$$\begin{aligned} MRS_{h,t}/\gamma^t &= L_{h,t}^{\chi}\left(C_{h,t}/\gamma^t - \varphi C_{h,t-1}/(\gamma \cdot \gamma^{t-1})\right) \\ \widehat{MRS}_{h,t} &= L_{h,t}^{\chi}\left(\hat{C}_{h,t} - \varphi/\gamma \hat{C}_{h,t-1}\right) \end{aligned}$$

We now log-linearize the variables and use the approximation $(1 + l_{h,t})^{\chi} \approx 1 + \chi l_{h,t}$

$$\bar{MRS}(1 + mrs_{h,t}) = \bar{L}^{\chi}\bar{C}(1 + \chi l_{h,t})(1 + c_{h,t} - \varphi/\gamma - \varphi/\gamma c_{h,t-1})$$

Using the fact that $\bar{MRS} = \bar{C}(1 - \varphi/\gamma)\bar{L}^{\chi}$, we get

$$(1 + mrs_{h,t})\left(1 - \frac{\varphi}{\gamma}\right) = 1 - \frac{\varphi}{\gamma} + \left(1 - \frac{\varphi}{\gamma}\right)\chi l_{h,t} + c_{h,t} - \frac{\varphi}{\gamma}c_{h,t-1}$$

We rearrange the terms to get

$$mrs_{h,t} = \left(\frac{1}{1 - \varphi/\gamma}\right)\left(c_{h,t} - \frac{\varphi}{\gamma}c_{h,t-1}\right) + \chi l_{h,t}$$

We finally integrate over h to get

$$mrs_t = \left(\frac{1}{1-\varphi/\gamma}\right)(c_t - \varphi/\gamma c_{t-1}) + \chi l_t \tag{G.18}$$

G.9 Resource constraint

The resource constraint is defined as follows

$$Y_t = C_t + I_t + a(U_t)K_t + G_t$$

Log-linearizing each term leads to

$$\bar{Y}(1+y_t) = \bar{C}(1+c_t) + \bar{I}(1+i_t) + \bar{K}\bar{R}^k u_t + \bar{G}(1+g_t)$$

We use the steady-state result that $\bar{Y} = \bar{C} + \bar{I} + \bar{G}$ and divide each side by $\bar{Y}$ to simplify the expression above to

$$y_t = c_y c_t + i_y i_t + \bar{R}^k k_y u_t + g_y g_t,$$

where $c_y = \bar{C}/\bar{Y}$, $i_y = \bar{I}/\bar{Y}$, $k_y = \bar{K}/\bar{Y}$ and $g_y = \bar{G}/\bar{Y}$. We now derive a formula for g_t by noticing that government expenditures follow the rule below

$$\frac{G_t}{Y_t} = g_y + x_t^g g_y(1 + g_t - y_t) = g_y + x_t^g$$
$$g_t = y_t + \frac{x_t^g}{g_y}$$

Plugging the result above into the log-linearized version of the resource constraint implies

$$y_t = \frac{1}{1-g_y}\left(\frac{\bar{C}}{\bar{Y}}c_t + \frac{\bar{I}}{\bar{Y}}i_t + \frac{\bar{R}^k\bar{K}}{\bar{Y}}u_t + x_t^g\right) \tag{G.19}$$

H Derivation of price- and wage-setting conditions

H.1 Cost minimization problem

$$\min_{\{K_{i,t},L_{i,t}\}} E_{i,t}\left[W_t L_{i,t} + R_t^k K_{i,t}\right] - \gamma^t \Phi_p$$
$$\text{s.t. } Y_{i,t} = A_t K_{i,t}^{\alpha} L_{i,t}^{1-\alpha},$$

where A_t follows

$$\log(A_t) = \rho_a \log(A_{t-1}) + \varepsilon_t^a \tag{H.1}$$

The lagrangean is:

$$\min_{\{K_{i,t},L_{i,t}\}} \mathcal{L} = E_{i,t}\left[W_t L_{i,t} + R_t^k K_{i,t} - \gamma^t \Phi_p + \mu_{i,t}\left[Y_{i,t} - A_t K_{i,t}^{\alpha} L_{i,t}^{1-\alpha}\right]\right]$$

The first-order conditions are:

$$E_{i,t}\left[W_t\right] = (1-\alpha)\mu_{i,t}\frac{E_{i,t}\left[Y_{i,t}\right]}{L_{i,t}}$$
$$E_{i,t}\left[R_t^k\right] = \alpha\mu_{i,t}\frac{E_{i,t}\left[Y_{i,t}\right]}{K_{i,t}}$$

Dividing both equations yields:

$$\frac{E_{i,t}\left[W_t\right]}{E_{i,t}\left[R_t^k\right]} = \frac{1-\alpha}{\alpha}\frac{K_{i,t}}{L_{i,t}} \tag{H.2}$$

substituting into the production function and solving for $K_{i,t}$ and $L_{i,t}$ as a function of $Y_{i,t}$, it follows that:

$$Y_{i,t} = A_t\left[\frac{\alpha}{1-\alpha}\frac{E_{i,t}\left[W_t\right]}{E_{i,t}\left[R_t^k\right]}\right]^{\alpha} L_{i,t} \qquad L_{i,t} = \left[\frac{1-\alpha}{\alpha}\frac{E_{i,t}\left[R_t^k\right]}{E_{i,t}\left[W_t\right]}\right]^{\alpha}\frac{Y_{i,t}}{A_t}$$
$$Y_{i,t} = A_t\left[\frac{1-\alpha}{\alpha}\frac{E_{i,t}\left[R_t^k\right]}{E_{i,t}\left[W_t\right]}\right]^{1-\alpha} K_{i,t} \qquad K_{i,t} = \left[\frac{\alpha}{1-\alpha}\frac{E_{i,t}\left[W_t\right]}{E_{i,t}\left[R_t^k\right]}\right]^{1-\alpha}\frac{Y_{i,t}}{A_t}$$

Substituting both into the total cost function, we get:

$$\begin{aligned}\mathcal{C}(Y_{i,t}) &= \left[W_t\left[\frac{1-\alpha}{\alpha}\frac{E_{i,t}\left[R_t^k\right]}{E_{i,t}\left[W_t\right]}\right]^{\alpha} + R_t^k\left[\frac{\alpha}{1-\alpha}\frac{E_{i,t}\left[W_t\right]}{E_{i,t}\left[R_t^k\right]}\right]^{1-\alpha}\right]\frac{Y_{i,t}}{A_t} - A_t\Phi_p \\ &= \left[\frac{(1-\alpha)W_t E_{i,t}\left[R_t^k\right] + \alpha R_t^k E_{i,t}\left[W_t\right]}{\alpha^{\alpha}(1-\alpha)^{1-\alpha}E_{i,t}\left[W_t\right]^{\alpha} E_{i,t}\left[R_t^k\right]^{1-\alpha}}\right]\frac{Y_{i,t}}{A_t} - A_t\Phi_p\end{aligned}$$

Finally, the marginal cost is given by:

$$MC_{i,t} \equiv \mathcal{C}'(Y_{i,t}) = \left[\frac{(1-\alpha)W_t E_{i,t}\left[R_t^k\right] + \alpha R_t^k E_{i,t}\left[W_t\right]}{\alpha^\alpha (1-\alpha)^{1-\alpha} E_{i,t}\left[W_t\right]^\alpha E_{i,t}\left[R_t^k\right]^{1-\alpha}} \right] \frac{1}{A_t} \tag{H.3}$$

H.2 Price index

The price index is given by

$$P_t = \left[\int_0^1 P_{i,t}^{-\frac{1}{\mu_t^p}} di \right]^{-\mu_t^p}$$

The firms that do not optimize follow the indexation rule given by

$$X_t = \Pi_{t-1}^{\iota_p} \bar{\Pi}^{1-\iota_p}$$

where $\iota_p \in [0,1]$.

Observe that

$$\begin{aligned} P_t^{-\frac{1}{\mu_t^p}} &= \int_0^1 P_{i,t}^{-\frac{1}{\mu_t^p}} di = \int_0^{\xi_p} (X_t P_{i,t-1})^{-\frac{1}{\mu_t^p}} di + \int_{\xi_p}^1 (P_{i,t}^*)^{-\frac{1}{\mu_t^p}} di \\ &= \xi_p X_t^{-\frac{1}{\mu_t^p}} \int_0^1 P_{i,t-1}^{-\frac{1}{\mu_t^p}} di + (1-\xi_p) \int_0^1 (P_{i,t}^*)^{-\frac{1}{\mu_t^p}} di \\ &= \xi_p (X_t P_{t-1})^{-\frac{1}{\mu_t^p}} + (1-\xi_p)(P_t^*)^{-\frac{1}{\mu_t^p}} \end{aligned}$$

where the last equality uses the definition of P_t in period $t-1$ and uses the definition of aggregate optimal price given by $P_t^* = \left[\int_0^1 (P_{i,t}^*)^{-\frac{1}{\mu_t^p}} di \right]^{-\mu_t^p}$.

Dividing both sides by $P_{t-1}^{-\frac{1}{\mu_t^p}}$ and using the definition of Π_t and X_t:

$$\Pi_t^{-\frac{1}{\mu_t^p}} = \xi_p \left(\Pi_{t-1}^{\iota_p} \bar{\Pi}^{1-\iota_p} \right)^{-\frac{1}{\mu_t^p}} + (1-\xi_p) \left(\hat{P}_t^* \right)^{-\frac{1}{\mu_t^p}} \tag{H.4}$$

where $\hat{P}_t^* = \frac{P_t^*}{P_{t-1}}$.

H.3 Optimal price

$$\max_{\{P^*_{i,t}\}} \quad \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s}\left(P^*_{i,t}X_{t,t+s}Y_{t+s|t} - \mathcal{C}(Y_{t+s|t})\right)\right]$$

$$\text{s.t.} \quad Y_{t+s|t} = \left(\frac{P^*_{i,t}X_{t,t+s}}{P_{t+s}}\right)^{-\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}} C_{t+s}$$

$$X_{t,t+s} = \begin{cases} \prod_{j=1}^{s}(\Pi_{t+j-1}^{\iota_p}\bar{\Pi}^{1-\iota_p}), & \text{for } s \geq 1 \\ 1, & \text{for } s = 0 \end{cases}$$

$$\Lambda_{t,t+s} = \beta^s \frac{\lambda_{t+s}}{\lambda_t} = \beta^s \frac{u_c(C_{t+s})}{u_c(C_t)} \frac{P_t}{P_{t+s}}$$

where $Y_{t+s|t}$ denote the output from period $t+s$ given that the last price update was done in period t.

FOC:

$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s}\left(X_{t,t+s}Y_{t+s|t} - \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right)P^*_{i,t}X_{t,t+s}\frac{Y_{t+s|t}}{P^*_{i,t}} + \mathcal{C}'(Y_{t+s|t})\left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right)\frac{Y_{t+s|t}}{P^*_{i,t}}\right)\right] = 0$$

$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s}Y_{t+s|t}\left(-\frac{X_{t,t+s}}{\lambda_{p,t+s}} + \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right)\frac{MC_{i,t+s}}{P^*_{i,t}}\right)\right] = 0 \tag{H.5}$$

$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s}Y_{t+s|t}\frac{X_{t,t+s}}{\lambda_{p,t+s}}\right] = \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s}Y_{t+s|t}\left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right)\frac{MC_{i,t+s}}{P^*_{i,t}}\right] \tag{H.6}$$

H.4 Wage-setting problem

$$\min_{\{W^*_{h,t}\}} \quad -E_{h,t}\left[\sum_{s=0}^{\infty}(\beta\xi_w)^s V\left(L_{h,t+s|t}\right)\right]$$

$$\text{s.t.} \quad L_{h,t+s|t} = \left(\frac{W^*_{h,t}X^w_{t,t+s}}{W_{t+s}}\right)^{-\frac{1+\mu_{w,t+s}}{\mu_{w,t+s}}} L_{t+s}$$

$$P_{t+s}C_{h,t+s} + P_{t+s}I_{h,t+s} + B_{h,t+s} + P_{t+s}a(U_{h,t+s})K_{h,t+s-1} + Q_{t+s+1,t}A_{h,t+s}$$
$$= R_{t+s-1}B_{h,t+s-1} + W_{h,t+s}L_{h,t+s|t} + R^k_{t+s}U_{h,t+s}K_{h,t+s-1} + P_{t+s}A_{h,t+s-1} + T_{h,t+s}$$

$$W_{h,t+s} = X^w_{t,t+s}W^*_{h,t}$$

$$X^w_{t,t+s} = \begin{cases} (\gamma\bar{\Pi}^{1-\iota_w})^s \prod_{j=1}^{s}\left(\Pi_{t+j-1}^{\iota_w}\right) & \text{if} \quad s \geq 1 \\ 1 & \text{if} \quad s = 0. \end{cases}$$

where $L_{h,t+s|t}$ denotes the labor supply from period $t+s$ given that the last wage update was done in period t.

The Lagrangean of this problem is given by

$$\mathcal{L} = \sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[V\left(L_{h,t+s|t}\right) + \Lambda_{h,t+s}\left(R_{t+s-1}B_{h,t+s-1} + X^w_{t,t+s}W^*_{h,t}L_{h,t+s|t} + R^k_{t+s}U_{h,t+s}K_{h,t-1} + \right.\right.$$
$$\left.\left. P_{t+s}A_{h,t+s-1} + T_{h,t+s} - P_{t+s}C_{h,t+s} - P_{t+s}I_{h,t+s} - B_{h,t+s} - P_{t+s}a(U_{h,t+s})K_{h,t+s-1} - Q_{t+s+1,t}A_{h,t+s}\right)\right]$$

Note that the Lagrangean multiplier of this problem is equal to the marginal utility of consumption, $\Lambda_{h,t+s}$.

The optimal condition for setting the wage of differentiated labor, $W^*_{h,t}$, is

$$0 = \sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[\left(\Lambda_{h,t+s}\left(X^w_{t,t+s}L_{h,t+s|t} - \frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}L_{h,t+s|t}X^w_{t,t+s}\right) + \frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}V'\left(L_{h,t+s}\right)\frac{L_{h,t+s|t}}{W^*_{h,t}}\right)\right]$$
$$0 = \sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\Lambda_{h,t+s}\left(\frac{X^w_{t,t+s}}{\mu^w_{t+s}} + \frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{V'\left(L_{h,t+s}\right)}{\Lambda_{h,t+s}W^*_{h,t}}\right)\right]$$

Using the FOC from households' problem and the definition of the rate of marginal substitution between labor and consumption, one can see that

$$\begin{aligned}\frac{V'\left(L_{h,t+s|t}\right)}{\Lambda_{h,t+s}} &= L^{\chi}_{h,t+s|t}(C_{h,t+s} - \varphi/\gamma C_{h,t+s-1})E_{h,t+s}[P_{t+s}] \\ &= MRS_{h,t+s|t}E_{h,t+s}[P_{t+s}].\end{aligned}$$

Thus, the wage-setting equation can be written as

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\Lambda_{h,t+s}\frac{X^w_{t,t+s}}{\mu^w_{t+s}}\right] = \sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\Lambda_{h,t+s}\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{MRS_{h,t+s|t}E_{h,t+s}[P_{t+s}]}{W^*_{h,t}}\right] \tag{H.7}$$

H.5 Labor packers

There is a continuum of households indexed by $h \in [0,1]$ who supply differentiated labor services that are imperfect substitutes for other households' labor services. As in Erceg et al. [2000], it is assumed that there are "employment agencies" that combine households' labor supply using the following aggregator

$$L_t = \left(\int_0^1 L_{h,t}^{\frac{1}{1+\mu^w_t}}\,dh\right)^{1+\mu^w_t}, \tag{H.8}$$

where μ_t^w denote the agency's wage mark-up such that $\log(1+\mu_t^w) = \log(1+\mu_{t-1}^w) + x_t^w$. In turn, x_t^w follows the process

$$x_t^w = \rho_w x_{t-1}^w + \varepsilon_t^w, \quad \varepsilon_t^w \sim \mathcal{N}(0, \sigma_w^2). \tag{H.9}$$

Agencies pay each household h their wage, $W_{h,t}$, and sell a homogeneous labor service to intermediate firms at a cost of W_t. Therefore, they maximize profits

$$W_t L_t - \int_0^1 W_{h,t} L_{h,t} dh \tag{H.10}$$

subject to the nominal wage index

$$W_t = \left(\int_0^1 (W_{h,t})^{-\frac{1}{\mu_t^w}} dh \right)^{-\mu_t^w}.$$

Thus, the demand for labor for each household h is given by

$$L_{h,t} = \left(\frac{W_{h,t}}{W_t} \right)^{-\frac{1+\mu_t^w}{\mu_t^w}} L_t, \tag{H.11}$$

where $W_t = \left(\int_0^1 (W_{h,t})^{-\frac{1}{\mu_t^w}} dh \right)^{-\mu_t^w}$ is the nominal wage index. Given the optimal prices and the indexation rule, the price level has the following law of motion

$$W_t = \left[\xi_w \left(\gamma \Pi_{t-1}^{\iota_w} \bar{\Pi}^{1-\iota_w} W_{t-1} \right)^{-\frac{1}{\mu_t^w}} + (1-\xi_w)(W_t^*)^{-\frac{1}{\mu_t^w}} \right]^{-\mu_t^w}. \tag{H.12}$$

where $W_t^* = \int_0^1 W_{h,t}^* dh$

I Log-linearization of price- and wage-setting conditions

Let $z_t \equiv \log(Z_t) - \log(\bar{Z})$ be the log-deviation from the steady-state for any relevant variable Z_t.

I.1 Marginal cost

Equation (H.2)

$$
\begin{aligned}
\frac{E_{i,t}\left[W_t\right]}{E_{i,t}\left[R_t^k\right]} &= \frac{1-\alpha}{\alpha}\frac{K_{i,t}}{L_{i,t}} \\
\frac{E_{i,t}\left[W_t/P_t\right]}{E_{i,t}\left[R_t^k/P_t\right]} &= \frac{1-\alpha}{\alpha}\frac{K_{i,t}}{L_{i,t}} \\
\frac{E_{i,t}\left[\hat{W}_t\right]}{E_{i,t}\left[\hat{R}_t^k\right]} &= \frac{1-\alpha}{\alpha}\frac{K_{i,t}}{L_{i,t}} \\
\frac{\bar{\hat{W}}}{\bar{\hat{R}}^k}\frac{E_{i,t}\left[e^{\hat{w}_t}\right]}{E_{i,t}\left[e^{\hat{r}_t^k}\right]} &= \frac{1-\alpha}{\alpha}\frac{\bar{K}}{\bar{L}}e^{k_{i,t}-l_{i,t}} \\
\frac{E_{i,t}\left[e^{\hat{w}_t}\right]}{E_{i,t}\left[e^{\hat{r}_t^k}\right]} &= e^{k_{i,t}-l_{i,t}}
\end{aligned}
$$

where the last equality uses the same equation in the steady state.

Log-linearizing each term:

$$
\begin{aligned}
\frac{E_{i,t}\left[1+\hat{w}_t\right]}{E_{i,t}\left[1+\hat{r}_t^k\right]} &= 1 + k_{i,t} - l_{i,t} \\
1 + E_{i,t}\left[\hat{w}_t\right] &= (1 + k_{i,t} - l_{i,t})(1 + E_{i,t}\left[\hat{r}_t^k\right]) \\
E_{i,t}\left[\hat{w}_t\right] &= k_{i,t} - l_{i,t} + E_{i,t}\left[\hat{r}_t^k\right]
\end{aligned}
$$

where last equality uses the fact that $x_t y_t \approxeq 0$ for any pair log-deviation from steady-state, (x_t, y_t).

Marginal cost (equation H.3)

$$MC_{i,t} = \left[\frac{(1-\alpha)W_t E_{i,t}\left[R_t^k\right] + \alpha R_t^k E_{i,t}\left[W_t\right]}{\alpha^\alpha(1-\alpha)^{1-\alpha}E_{i,t}\left[W_t\right]^\alpha E_{i,t}\left[R_t^k\right]^{1-\alpha}}\right]\frac{1}{A_t}$$

$$MC_{i,t}/P_t = \left[\frac{(1-\alpha)(W_t/P_t)E_{i,t}\left[R_t^k/P_t\right] + \alpha(R_t^k/P_t)E_{i,t}\left[W_t/P_t\right]}{\alpha^\alpha(1-\alpha)^{1-\alpha}E_{i,t}\left[W_t/P_t\right]^\alpha E_{i,t}\left[R_t^k/P_t\right]^{1-\alpha}}\right]\frac{1}{A_t}$$

$$\hat{MC}_{i,t} = \left[\frac{(1-\alpha)\hat{W}_t E_{i,t}\left[\hat{R}_t^k\right] + \alpha \hat{R}_t^k E_{i,t}\left[\hat{W}_t\right]}{\alpha^\alpha(1-\alpha)^{1-\alpha}E_{i,t}\left[\hat{W}_t\right]^\alpha E_{i,t}\left[\hat{R}_t^k\right]^{1-\alpha}}\right]\frac{1}{A_t}$$

$$\bar{MC}exp(mc_{i,t}) = \left[\frac{(1-\alpha)\bar{\hat{W}}\bar{\hat{R}}^k exp(\hat{w}_t + E_{i,t}\left[\hat{r}_t^k\right]) + \alpha\bar{\hat{W}}\bar{\hat{R}}^k exp(\hat{r}_t^k + E_{i,t}\left[\hat{w}_t\right])}{\alpha^\alpha(1-\alpha)^{1-\alpha}\bar{\hat{W}}^\alpha(\bar{\hat{R}}^k)^{1-\alpha}exp(\alpha E_{i,t}\left[\hat{w}_t\right] + (1-\alpha)E_{i,t}\left[\hat{r}_t^k\right])}\right]\frac{1}{exp(a_t)}$$

$$exp(mc_{it} + a_t) = \left[\frac{(1-\alpha)exp(\hat{w}_t + E_{i,t}\left[\hat{r}_t^k\right]) + \alpha exp(\hat{r}_t^k + E_{i,t}\left[\hat{w}_t\right])}{exp(\alpha E_{i,t}\left[\hat{w}_t\right] + (1-\alpha)E_{i,t}\left[\hat{r}_t^k\right])}\right]$$

where last equality uses that $\bar{MC} = \frac{\bar{\hat{W}}^{1-\alpha}(\bar{\hat{R}}^k)^\alpha}{\alpha^\alpha(1-\alpha)^{1-\alpha}}$

Log-linearizing each term:

$$1 + mc_{i,t} + a_t = \frac{(1-\alpha)(1 + \hat{w}_t + E_{i,t}\left[\hat{r}_t^k\right]) + \alpha(1 + \hat{r}_t^k + E_{i,t}\left[\hat{w}_t\right])}{1 + \alpha E_{i,t}\left[\hat{w}_t\right] + (1-\alpha)E_{i,t}\left[\hat{r}_t^k\right]}$$

$$(1 + mc_{i,t} + a_t)(1 + \alpha E_{i,t}\left[\hat{w}_t\right] + (1-\alpha)E_{i,t}\left[\hat{r}_t^k\right]) = (1-\alpha)(1 + \hat{w}_t + E_{i,t}\left[\hat{r}_t^k\right]) + \alpha(1 + \hat{r}_t^k + E_{i,t}\left[\hat{w}_t\right])$$

$$1 + mc_{i,t} + a_t + \alpha E_{i,t}\left[\hat{w}_t\right] + (1-\alpha)E_{i,t}\left[\hat{r}_t^k\right] = (1-\alpha)(1 + \hat{w}_t + E_{i,t}\left[\hat{r}_t^k\right]) + \alpha(1 + \hat{r}_t^k + E_{i,t}\left[\hat{w}_t\right])$$

$$mc_{i,t} = (1-\alpha)\hat{w}_t + \alpha\hat{r}_t^k - a_t$$

Note that $mc_{i,t}$ does not depend on i. This is due to the fact that the production function exhibits constant returns to scale, and, as it turns out, the heterogeneity of expectations does not play a role in the cost minimization problem up to the first-order approximation. Thus, we can simply drop the i such that

$$mc_t = (1-\alpha)\hat{w}_t + \alpha\hat{r}_t^k - a_t$$

I.2 Production function of intermediate producers

Production function:

$$Y_{i,t} = e^{x_t^a}K_{i,t}^\alpha\left(\gamma^t L_{i,t}\right)^{1-\alpha} - \gamma^t\Phi_p$$

We first detrend the variables to get

$$\hat{Y}_{i,t} = e^{x_t^a} \hat{K}_{i,t}^{\alpha} L_{i,t}^{1-\alpha} - \Phi_p$$

We move the term Φ_p to the left-hand side of the equation and log-linearize it as follows

$$\log(\hat{Y}_{i,t} + \Phi_p) - \log(\bar{Y}_i + \Phi_p) = x_t^a + \alpha k_{i,t} + (1-\alpha) l_{i,t}$$

We may rearrange the left-hand side of the expression above to get

$$\log(\hat{Y}_{i,t} + \Phi_p) - \log(\bar{Y}_i + \Phi_p) = \frac{\bar{Y}_i}{\bar{Y}_i + \Phi_p} y_{i,t}$$

We conclude that

$$y_{i,t} = \frac{\bar{Y}_i + \Phi_p}{\bar{Y}_i} \left(x_t^a + \alpha k_{i,t} + (1-\alpha) l_{i,t} \right)$$

Integrating over i, we finally get

$$y_t = \frac{\bar{Y} + \Phi_p}{\bar{Y}} \left(x_t^a + \alpha k_t + (1-\alpha) l_t \right) \tag{I.1}$$

I.3 Price index

Price index (equation H.4) requires a different log-linearization rule. Consider $X_t = Y_t^{Z_t}$. Then, $x_t = \bar{Z}(\log(\bar{Y}) z_t + y_t)$ Applying this to equation (H.4):

$$\Pi_t^{-\frac{1}{\mu_t^p}} = \xi_p \left(\Pi_{t-1}^{\iota_p} \bar{\Pi}^{1-\iota_p} \right)^{-\frac{1}{\mu_t^p}} + (1-\xi_p) \left(\hat{P}_t^* \right)^{-\frac{1}{\mu_t^p}}$$

$$-\frac{1}{\bar{\lambda}_p}(-\log(\bar{\Pi})\mu_t^p + \pi_t) = -\xi_p \left(\frac{1}{\bar{\lambda}_p}(-\log(\bar{\Pi})\mu_t^p + \iota_p \pi_{t-1}) \right) - (1-\xi_p) \frac{1}{\bar{\lambda}_p}(-\log(\bar{\Pi})\mu_t^p + \hat{p}_t^*),$$

which rearranging leads to

$$\pi_t = \xi_p \iota_p \pi_{t-1} + (1-\xi_p) \hat{p}_t^* \tag{I.2}$$

I.4 Optimal price

Optimal price (equation H.6):

$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \frac{X_{t,t+s}}{\lambda_{p,t+s}}\right] = \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right) \frac{MC_{i,t+s}}{P_{i,t}^*}\right]$$
$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \frac{X_{t,t+s}}{\lambda_{p,t+s}}\right] = \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right) \frac{MC_{i,t+s}}{P_{t+s}} \frac{P_{t-1}}{P_{i,t}^*} \frac{P_{t+s}}{P_{t-1}}\right]$$
$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \frac{X_{t,t+s}}{\lambda_{p,t+s}}\right] = \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right) \frac{\hat{MC}_{i,t+s}\Pi_{t-1,t+s}}{\hat{P}_{i,t}^*}\right]$$

where $\Pi_{t-1,t+s} = \frac{P_{t+s}}{P_{t-1}} = \sum_{j=0}^{s} \Pi_{t+j}$ and $\hat{P}_{i,t}^* = P_{i,t}^*/P_{t-1}$. If we stationarize $\hat{P}_{i,t}^*$ by P_t, we wouldn't be able to take it as known by firms, since P_t is unknown. Thus, we proceed by dividing by P_{t-1}.

The equation above can be rewritten as

$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \frac{X_{t,t+s}}{\lambda_{p,t+s}}\right] = \sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\Lambda_{t,t+s} Y_{t+s|t} \left(\frac{1+\lambda_{p,t+s}}{\lambda_{p,t+s}}\right) \frac{\hat{MC}_{i,t+s}\Pi_{t-1,t+s}}{\hat{P}_{i,t}^*}\right]$$
$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\frac{\bar{\Lambda}_s \bar{Y} \bar{X}_s}{\bar{\lambda}_p} exp\left(\lambda_{t,t+s} + y_{t+s|t} + x_{t,t+s} - \tilde{\lambda}_{p,t+s}\right)\right] =$$
$$\sum_{s=0}^{\infty} \xi_p^s E_{i,t}\left[\frac{\bar{\Lambda}_s \bar{Y}(1+\bar{\lambda}_p)\bar{MC}\bar{\Pi}^{s+1}}{\bar{\lambda}_p \bar{\Pi}} exp\left(\lambda_{t,t+s} + y_{t+s|t} + x_{t,t+s} + \tilde{\lambda}_{p,t+s} - \tilde{\lambda}_{p,t+s} + \hat{mc}_{i,t+s} + \pi_{t-1,t+s} - \hat{p}_{i,t}^*\right)\right]$$

where $x_t^p = \log(\mu_t^p) - \log(\bar{\lambda}_p) \approx \log(1+\mu_t^p) - \log(1+\bar{\lambda}_p)$.

Note in the steady-state $\bar{\Lambda}_s = \left(\frac{\beta}{\bar{\Pi}}\right)^s$, $\bar{X}_s = \prod_{j=1}^{s}(\bar{\Pi}) = \bar{\Pi}^s$ and $\bar{MC} = 1/(1+\bar{\lambda}_p)$. Thus, the equation above can be written as

$$\sum_{s=0}^{\infty} (\beta\xi_p)^s E_{i,t}\left[exp\left(\lambda_{t,t+s} + y_{t+s|t} + x_{t,t+s} - \tilde{\lambda}_{p,t+s}\right)\right] =$$
$$\sum_{s=0}^{\infty} (\beta\xi_p)^s E_{i,t}\left[exp\left(\lambda_{t,t+s} + y_{t+s|t} + \hat{mc}_{i,t+s} + \pi_{t-1,t+s} - \hat{p}_{i,t}^*\right)\right]$$

Loglinearizing leads to:

$$\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[1+\lambda_{t,t+s}+y_{t+s|t}+x_{t,t+s}-\tilde{\lambda}_{p,t+s}\right]=\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}[1+\lambda_{t,t+s}+y_{t+s|t}+\hat{mc}_{i,t+s}$$
$$+\pi_{t-1,t+s}-\hat{p}^*_{i,t}]$$
$$\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[x_{t,t+s}-\tilde{\lambda}_{p,t+s}\right]=\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s}+\pi_{t-1,t+s}-\hat{p}^*_{i,t}\right]$$
$$\frac{\hat{p}^*_{i,t}}{1-\beta\xi_p}=\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s}+\pi_{t-1,t+s}-x_{t,t+s}+\tilde{\lambda}_{p,t+s}\right]$$
$$\hat{p}^*_{i,t}=(1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s}+\pi_{t-1,t+s}-x_{t,t+s}+\tilde{\lambda}_{p,t+s}\right]$$

Note that for $s \geq 1$, we have that

$$X_{t,t+s}=\prod_{j=1}^{s}(\Pi^{\iota_p}_{t+j-1}\bar{\Pi}^{1-\iota_p})$$
$$\bar{\Pi}^s exp(x_{t,t+s})=\bar{\Pi}^s exp\left(\iota_p\sum_{j=1}^{s}\pi_{t+j-1}\right)$$

Log-linearizing, we find that:

$$x_{t,t+s}=\iota_p\sum_{j=1}^{s}\pi_{t+j-1} \tag{I.3}$$

and $x_{t,t}=0$ by the definition of $X_{t,t+s}$. Analogously, we have that

$$\pi_{t-1,t+s}=\sum_{j=0}^{s}\pi_{t+j} \tag{I.4}$$

Using equations (I.3-I.4) and the fact that the marginal cost does not depend on i, the optimal price is given by

$$\hat{p}^*_{i,t}=(1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{t+s}+\sum_{j=0}^{s}\pi_{t+j}-\iota_p\sum_{j=1}^{s}\pi_{t+j-1}+\tilde{\lambda}_{p,t+s}\right]$$

We now use the usual trick to rewrite $\hat{p}^*_{i,t}$ recursively.

$$
\begin{aligned}
\hat{p}^*_{i,t} &= (1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{t+s} + \sum_{j=0}^{s}\pi_{t+j} - \iota_p\sum_{j=1}^{s}\pi_{t+j-1} + \tilde{\lambda}_{p,t+s}\right] \\
&= (1-\beta\xi_p)E_{i,t}\left[\hat{mc}_t + \pi_t + x^p_t\right] \\
&+ (1-\beta\xi_p)\sum_{s=1}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{t+s} + \sum_{j=0}^{s}\pi_{t+j} - \iota_p\sum_{j=1}^{s}\pi_{t+j-1} + \tilde{\lambda}_{p,t+s}\right] \\
&= (1-\beta\xi_p)E_{i,t}\left[\hat{mc}_t + \pi_t + x^p_t\right] \\
&+ (1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^{s+1} E_{i,t}\left[\hat{mc}_{i,t+s+1} + \sum_{j=0}^{s+1}\pi_{t+j} - \iota_p\sum_{j=1}^{s+1}\pi_{t+j-1} + \tilde{\lambda}_{p,t+s+1}\right] \\
&= (1-\beta\xi_p)E_{i,t}\left[\hat{mc}_t + \pi_t + x^p_t\right] \\
&+ (1-\beta\xi_p)\beta\xi_p\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s+1} + \pi_t + \sum_{j=1}^{s+1}\pi_{t+j} - \iota_p\left(\pi_t + \sum_{j=2}^{s+1}\pi_{t+j-1}\right) + \tilde{\lambda}_{p,t+s+1}\right] \\
&= (1-\beta\xi_p)E_{i,t}\left[\hat{mc}_t + \pi_t + x^p_t\right] + \beta\xi_p(1-\iota_p)E_{i,t}\left[\pi_t\right] \\
&+ (1-\beta\xi_p)\beta\xi_p\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s+1} + \sum_{j=0}^{s}\pi_{t+j+1} - \iota_p\sum_{j=1}^{s}\pi_{t+j} + \tilde{\lambda}_{p,t+s+1}\right]
\end{aligned}
$$

where the second equality separates the $s = 0$ term from the summation, and the third redefines the summation counter as $s = s - 1$. The fourth equality separates the first term from the summations inside the brackets, and the fifth redefines the summation counter as $j = j - 1$ and solves the infinite sum for terms that do not depend on s.

Taking the expectation of the optimal price one period ahead leads to

$$
\begin{aligned}
E_{i,t}[\hat{p}^*_{i,t+1}] &= (1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[E_{i,t+1}\left[\hat{mc}_{i,t+s+1} + \sum_{j=0}^{s}\pi_{t+j+1} - \iota_p\sum_{j=1}^{s}\pi_{t+j} + \tilde{\lambda}_{p,t+s+1}\right]\right] \\
&= (1-\beta\xi_p)\sum_{s=0}^{\infty}(\beta\xi_p)^s E_{i,t}\left[\hat{mc}_{i,t+s+1} + \sum_{j=0}^{s}\pi_{t+j+1} - \iota_p\sum_{j=1}^{s}\pi_{t+j} + \tilde{\lambda}_{p,t+s+1}\right]
\end{aligned}
$$

using the law of iterated expectations (which holds for i-th expectation but not for the average expectation). Thus, the optimal price can be written as

$$
\hat{p}^*_{i,t} = (1-\beta\xi_p)E_{i,t}\left[\hat{mc}_t + x^p_t\right] + (1-\iota_p\beta\xi_p)E_{i,t}\left[\pi_t\right] + \beta\xi_p E_{i,t}\left[\hat{p}^*_{i,t+1}\right] \tag{I.5}
$$

Finally, we need to aggregate the optimal price to find the NKPC. Aggregating $\hat{p}^*_{i,t}$ leads

to

$$\hat{p}_t^* = (1-\beta\xi_p)\bar{E}_t\left[\hat{mc}_t + x_t^p\right] + (1-\iota_p\beta\xi_p)\bar{E}_t\left[\pi_t\right] + \beta\xi_p\int_0^1 E_{i,t}\left[\hat{p}_{i,t+1}^*\right]di$$

Note that $\int_0^1 E_{i,t}\left[\hat{p}_{i,t+1}^*\right]di \neq \bar{E}_t\left[\hat{p}_{t+1}^*\right]$.

Substituting into equation (I.2) leads to the NKPC under heterogeneous information and indexation

$$\pi_t = \xi_p\iota_p\pi_{t-1} + \kappa_p\xi_p\bar{E}_t\left[\hat{mc}_t + x_t^p\right] + (1-\xi_p)(1-\iota_p\beta\xi_p)\bar{E}_t\left[\pi_t\right] + (1-\xi_p)\beta\xi_p\int_0^1 E_{i,t}\left[\hat{p}_{i,t+1}^*\right]di \tag{I.6}$$

where $\kappa_p = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}$

I.5 Wage index

We can rewrite the wage index $W_t = \left(\int_0^1 (W_{h,t})^{-\frac{1}{\mu_t^w}}dh\right)^{-\mu_t^w}$ as follows

$$\begin{aligned}
W_t &= \left[\xi_w\left(\gamma\Pi_{t-1}^{\iota_w}\bar{\Pi}^{1-\iota_w}W_{t-1}\right)^{-\frac{1}{\mu_t^w}} + (1-\xi_w)\left(W_t^*\right)^{-\frac{1}{\mu_t^w}}\right]^{-\mu_t^w} \\
\frac{W_t}{\gamma^t P_t} &= \left[\xi_w\left(\Pi_{t-1}^{\iota_w}\bar{\Pi}^{1-\iota_w}\frac{W_{t-1}}{\gamma^{t-1}P_{t-1}}\frac{P_{t-1}}{P_t}\right)^{-\frac{1}{\mu_t^w}} + (1-\xi_w)\left(\frac{W_t^*}{\gamma^t P_{t-1}}\frac{P_{t-1}}{P_t}\right)^{-\frac{1}{\mu_t^w}}\right]^{-\mu_t^w} \\
(\hat{W}_t\Pi_t)^{-\frac{1}{\mu_t^w}} &= \xi_w\left(\Pi_{t-1}^{\iota_w}\bar{\Pi}^{1-\iota_w}\hat{W}_{t-1}\right)^{-\frac{1}{\mu_t^w}} + (1-\xi_w)(\hat{W}_t^*)^{-\frac{1}{\mu_t^w}}
\end{aligned}$$

Note that in the steady-state: $(\bar{\hat{W}}\bar{\Pi})^{-\frac{1}{\bar{\mu}^w}} = \xi_w\left(\bar{\Pi}\bar{\hat{W}}\right)^{-\frac{1}{\bar{\mu}^w}} + (1-\xi_w)\left(\bar{\hat{W}}^*\right)^{-\frac{1}{\bar{\mu}^w}}$, which implies that $\bar{\hat{W}}^* = \bar{\hat{W}}\bar{\Pi}$.

Following the same log-linearization technique from the price index leads to

$$\begin{aligned}
-\frac{1}{\bar{\mu}^w}(-\log(\bar{\Pi}\bar{\hat{W}})\mu_t^w + \hat{w}_t + \pi_t) = &-\xi_w\left(\frac{1}{\bar{\mu}^w}(-\log(\bar{\Pi}\bar{\hat{W}})\mu_t^w + \iota_w\pi_{t-1} + \hat{w}_{t-1})\right) \\
&- (1-\xi_w)\frac{1}{\bar{\mu}^w}(-\log(\bar{\hat{W}}^*)\mu_t^w + \hat{w}_t^*),
\end{aligned}$$

which rearranging leads to

$$\hat{w}_t + \pi_t = \xi_w(\iota_w\pi_{t-1} + \hat{w}_{t-1}) + (1-\xi_w)\hat{w}_t^* \tag{I.7}$$

where $\hat{w}_t^* = \int_0^1 \hat{w}_{h,t}^* dh$.

I.6 Optimal wage

The optimal condition of the wage setting problem is

$$E_{h,t}\left[\sum_{s=0}^{\infty}(\beta\xi_w)^s\left(\Lambda_{t,t+s}\frac{X^w_{t,t+s}}{\mu^w_{t+s}}-\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{L^\chi_{h,t+s}}{W^*_{h,t}}\right)L_{h,t+s}\right]=0 \tag{I.8}$$

It follows from equation (I.8) that

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\Lambda_{h,t+s}\frac{X^w_{t,t+s}}{\mu^w_{t+s}}\right]=\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\Lambda_{h,t+s}\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{MRS_{h,t+s|t}E_{h,t+s}[P_{t+s}]}{W^*_{h,t}}\right]$$

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\frac{\Lambda_{h,t+s}\gamma^{t+s}P_{t-1}}{\gamma^{t+s}P_{t+s}}\frac{X^w_{t,t+s}}{\mu^w_{t+s}}\right]$$

$$=\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\frac{\Lambda_{h,t+s}\gamma^{t+s}P_{t-1}}{\gamma^{t+s}P_{t+s}}\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{MRS_{h,t+s|t}P_{t-1}\gamma^t E_{h,t+s}[P_{t+s}]}{\gamma^t W^*_{h,t}P_{t-1}}\right]$$

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\frac{\hat{\Lambda}_{h,t+s}}{\gamma^s}\frac{X^w_{t,t+s}}{\mu^w_{t+s}\Pi_{t-1,t+s}}\right]$$

$$=\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[L_{h,t+s|t}\hat{\Lambda}_{h,t+s}\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}\frac{M\hat{R}S_{h,t+s|t}E_{h,t+s}[\Pi_{t-1,t+s}]}{\hat{W}^*_{h,t}\Pi_{t-1,t+s}}\right]$$

This equation in the steady state implies that

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s\bar{L}\frac{\bar{\hat{\Lambda}}}{\gamma^s}\frac{(\gamma\bar{\Pi})^s}{\bar{\mu}^w\bar{\Pi}^s}=\sum_{s=0}^{\infty}(\beta\xi_w)^s\bar{L}\bar{\hat{\Lambda}}\frac{1+\bar{\mu}^w}{\bar{\mu}^w}\frac{M\bar{\hat{R}}S\bar{\Pi}^{s+1}}{\bar{\hat{W}}\bar{\Pi}^{s+1}}$$

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s=\sum_{s=0}^{\infty}(\beta\xi_w)^s\frac{(1+\bar{\mu}^w)M\bar{\hat{R}}S}{\bar{\hat{W}}}$$

$$\bar{\hat{W}}=(1+\bar{\mu}^w)M\bar{\hat{R}}S$$

Now we turn to the log-linearization:

$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[exp\left(l_{h,t+s|t}+\hat{\lambda}_{h,t+s}+x^w_{t,t+s}-\pi_{t-1,t+s}-\tilde{\mu}^w_{t+s}\right)\right]$$
$$=\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[exp\left(l_{h,t+s|t}+\hat{\lambda}_{h,t+s}+\hat{mrs}_{h,t+s|t}+E_{h,t+s}[\pi_{t-1,t+s}]-\hat{w}^*_{h,t}-\pi_{t-1,t+s}\right)\right]$$
$$\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[1+l_{h,t+s|t}+\hat{\lambda}_{h,t+s}+x^w_{t,t+s}-\pi_{t-1,t+s}-\tilde{\mu}^w_{t+s}\right]$$
$$=\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[1+l_{h,t+s|t}+\hat{\lambda}_{h,t+s}+\hat{mrs}_{h,t+s|t}-\hat{w}^*_{h,t}\right]$$
$$\hat{w}^*_{h,t}=(1-\beta\xi_w)\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[\hat{mrs}_{h,t+s|t}+\pi_{t-1,t+s}-x^w_{t,t+s}+\tilde{\mu}^w_{t+s}\right]$$
$$\hat{w}^*_{h,t}=(1-\beta\xi_w)\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[\hat{mrs}_{h,t+s|t}+\sum_{j=0}^{s}\pi_{t+j}-\iota_w\sum_{j=1}^{s}\pi_{t+j-1}+\tilde{\mu}^w_{t+s}\right]$$

where $\tilde{\mu}^w_t=\log(\mu^w_t)-\log(\bar{\mu}^w)\approx\log(1+\mu^w_t)-\log(1+\bar{\mu}^w)$.

Using the labor demand (14) at period $t+s$ for a given wage set at t, we have that

$$L_{h,t+s|t}=\left(\frac{W^*_{h,t}X^w_{t,t+s}}{W_{t+s}}\right)^{-\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}}L_{t+s}$$
$$L_{h,t+s|t}=\left(\frac{\hat{W}^*_{h,t}}{\gamma^{t+s}P_{t-1}}\frac{\gamma^{t+s}P_{t+s}}{\hat{W}_{t+s}}\frac{P_{t-1}}{P_{t+s}}X^w_{t,t+s}\right)^{-\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}}L_{t+s}$$
$$L_{h,t+s|t}=\left(\frac{\hat{W}^*_{h,t}X^w_{t,t+s}}{\gamma^s\hat{W}_{t+s}\Pi_{t-1,t+s}}\right)^{-\frac{1+\mu^w_{t+s}}{\mu^w_{t+s}}}L_{t+s}$$

Since $\bar{X}^w_s=(\gamma\bar{\Pi})^s$, $\bar{\Pi}_{-1,s}=\bar{\Pi}^{s+1}$, and $\bar{\hat{W}}^*=\bar{\hat{W}}\bar{\Pi}$, we have that $\bar{L}_h=\bar{L}$. Log-linearizing implies

$$l_{h,t+s|t}=-\frac{1+\bar{\mu}^w}{\bar{\mu}^w}\left[\hat{w}^*_{h,t}+\iota_w\sum_{j=1}^{s}\pi_{t+j-1}-\hat{w}_{t+s}-\sum_{j=0}^{s}\pi_{t+j}\right]+l_{t+s} \tag{I.9}$$

where the first equality uses that $\log\left(\frac{\bar{\hat{W}}^*\bar{X}^w_s}{\bar{\hat{W}}\gamma^s\bar{\Pi}^{s+1}}\right)=\log(1)=0$. and the second uses that, analogously to the indexation scheme for prices, $x^w_{t,t+s}=\iota_w\sum_{j=1}^{s}\pi_{t+j-1}$.

The MRS at period $t+s$ for a given wage set at t is given by

$$\begin{aligned}
\hat{mrs}_{h,t+s|t} &= \left(\frac{1}{1-\varphi/\gamma}\right)(c_{h,t+s} - \varphi/\gamma c_{h,t+s-1}) + \chi l_{t+s|t} \\
&= \left(\frac{1}{1-\varphi/\gamma}\right)(c_{h,t+s} - \varphi/\gamma c_{h,t+s-1}) \\
&+ \chi\left(-\frac{1+\bar{\mu}^w}{\bar{\mu}^w}\left[\hat{w}^*_{h,t} + \iota_w \sum_{j=1}^{s}\pi_{t+j-1} - \hat{w}_{t+s} - \sum_{j=0}^{s}\pi_{t+j}\right] + l_{t+s}\right)
\end{aligned}$$

Using the fact that $c_{h,t} = c_t$ for all h and the average MSR, mrs_t, we have that

$$\hat{mrs}_{h,t+s|t} = \hat{mrs}_{t+s} - \chi\theta_w\left[\hat{w}^*_{h,t} + \iota_w \sum_{j=1}^{s}\pi_{t+j-1} - \hat{w}_{t+s} - \sum_{j=0}^{s}\pi_{t+j}\right]$$

where $\theta_w \equiv \frac{1+\bar{\mu}^w}{\bar{\mu}^w}$.

Substituting it back into the optimal wage

$$\hat{w}^*_{h,t} = \frac{(1-\beta\xi_w)}{1+\chi\theta_w}\sum_{s=0}^{\infty}(\beta\xi_w)^s E_{h,t}\left[\hat{mrs}_{t+s} + \chi\theta_w\hat{w}_{t+s} + (1+\chi\theta_w)\left(\sum_{j=0}^{s}\pi_{t+j} - \iota_w\sum_{j=1}^{s}\pi_{t+j-1}\right) + \tilde{\mu}^w_{t+s}\right]$$

Following similar steps to the optimal price equation, we can rewrite the optimal wage as

$$\begin{aligned}
\hat{w}^*_{h,t} &= \frac{(1-\beta\xi_w)}{1+\chi\theta_w}E_{h,t}\left[(\hat{mrs}_t + \chi\theta_w w_t + (1+\chi\theta_w)\pi_t + \tilde{\mu}^w_t)\right] + (1-\iota_w)\beta\xi_w E_{h,t}\left[\pi_t\right] + \beta\xi_w E_{h,t}\left[\hat{w}^*_{h,t+1}\right] \\
&= \frac{(1-\beta\xi_w)}{1+\chi\theta_w}E_{h,t}\left[(\hat{mrs}_t + \chi\theta_w w_t + \tilde{\mu}^w_t)\right] + (1-\iota_w\beta\xi_w)E_{h,t}\left[\pi_t\right] + \beta\xi_w E_{h,t}\left[\hat{w}^*_{h,t+1}\right]
\end{aligned}$$

Integrating over h:

$$\hat{w}^*_t = \frac{(1-\beta\xi_w)}{1+\chi\theta_w}\bar{E}_t\left[(\hat{mrs}_t + \chi\theta_w w_t + \tilde{\mu}^w_t)\right] + (1-\iota_w\beta\xi_w)\bar{E}_t\left[\pi_t\right] + \beta\xi_w\int_0^1 E_{h,t}\left[\hat{w}^*_{h,t+1}\right]dh \tag{I.10}$$

J Impulse response functions

In this section, the remaining impulse responses for the FI and DI models estimated with the dataset including expectation data are displayed.

Figure J1: Impulse responses to preference shock

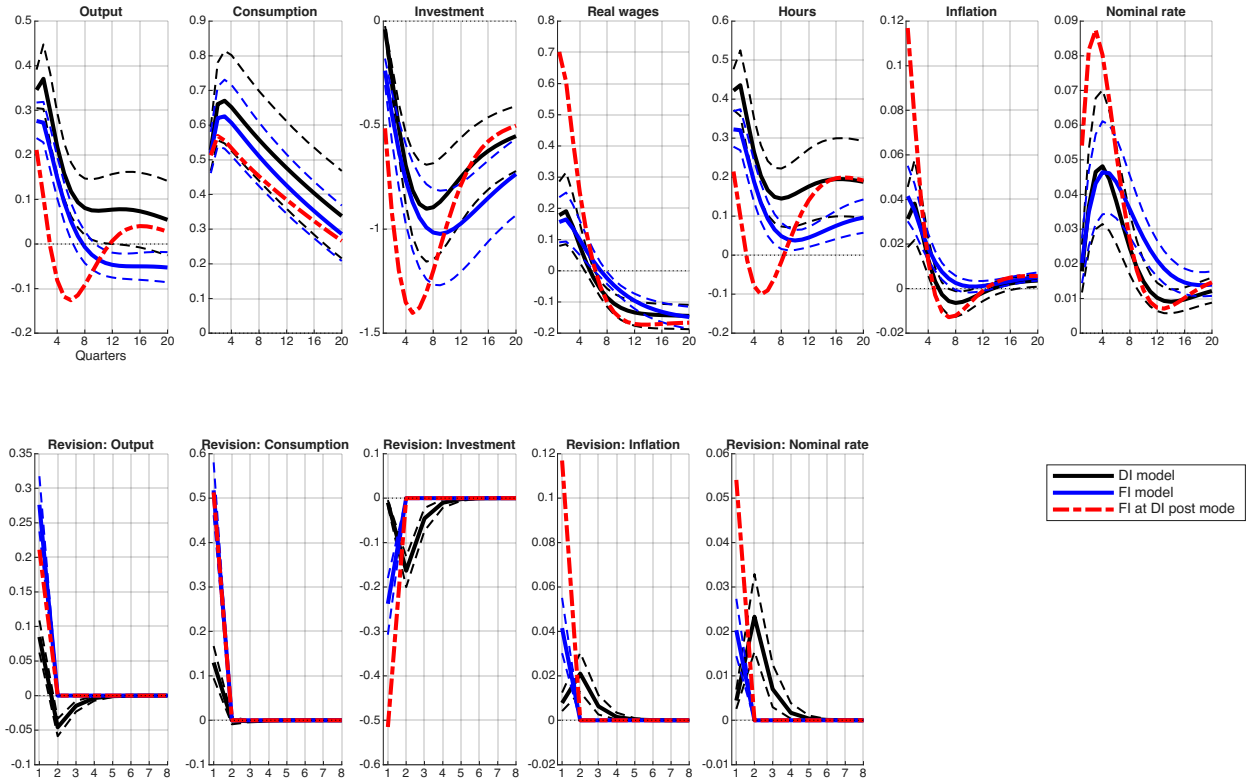


Figure J2: Impulse responses to TFP shock

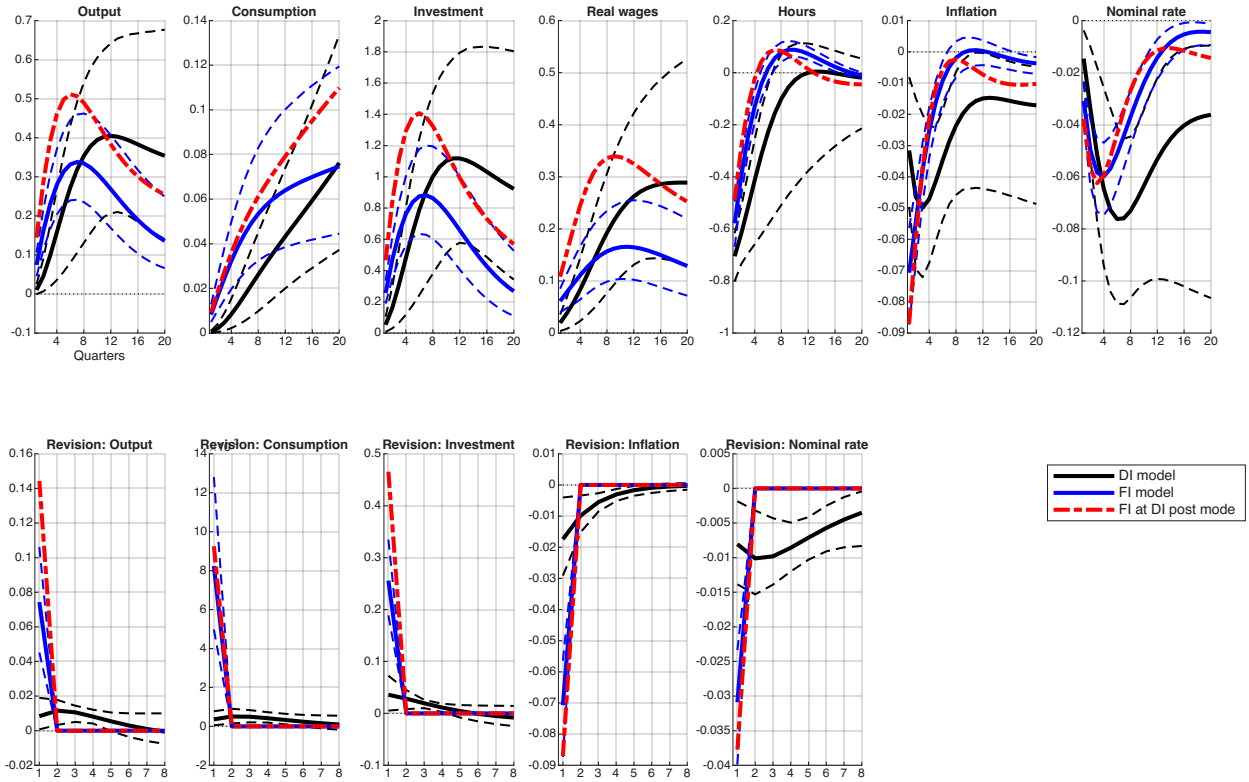


Figure J3: Impulse responses to wage mark-up shock

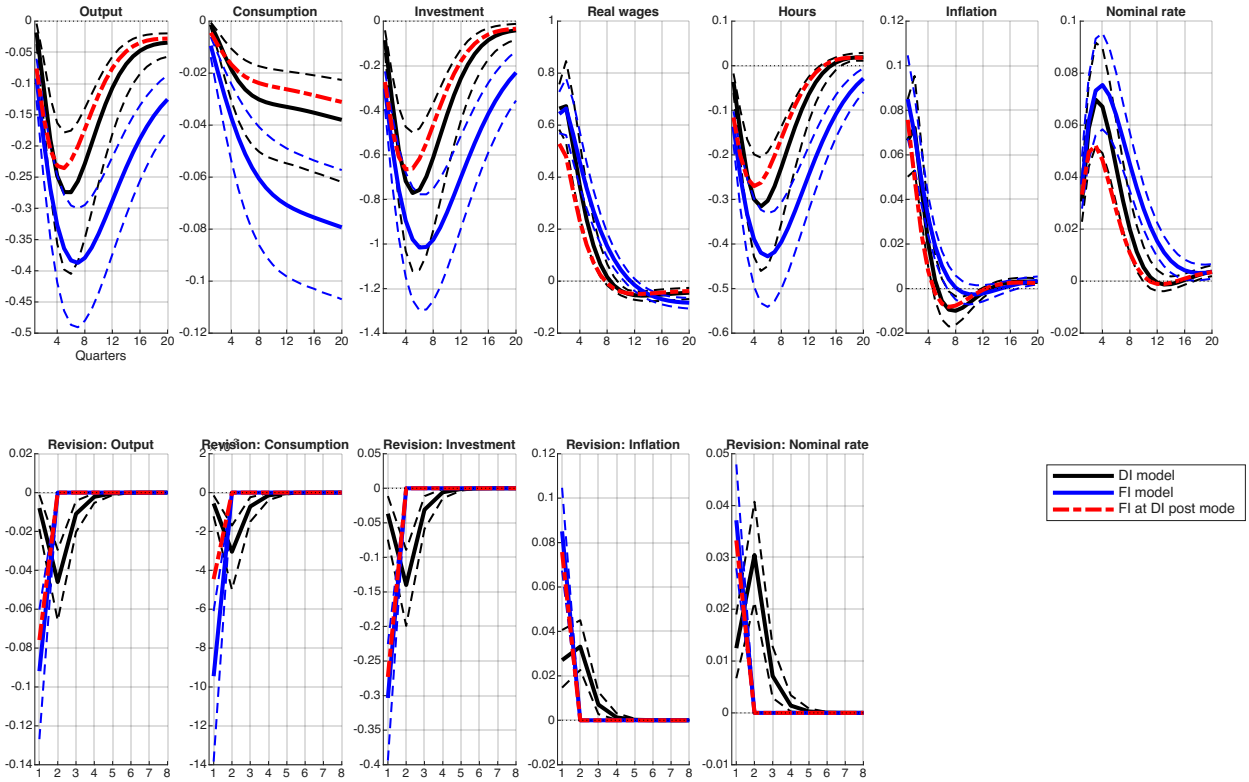


Figure J4: Impulse responses to price mark-up shock

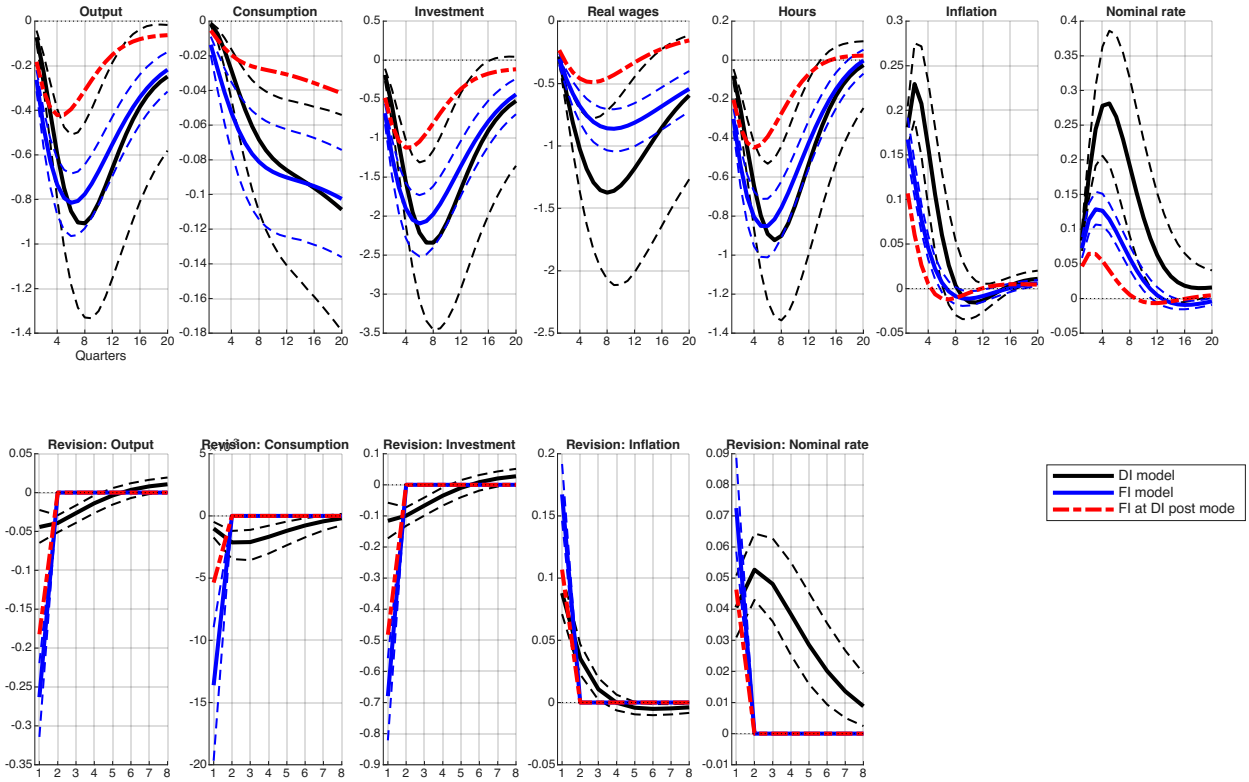


Figure J5: Impulse responses to government expenditure shock

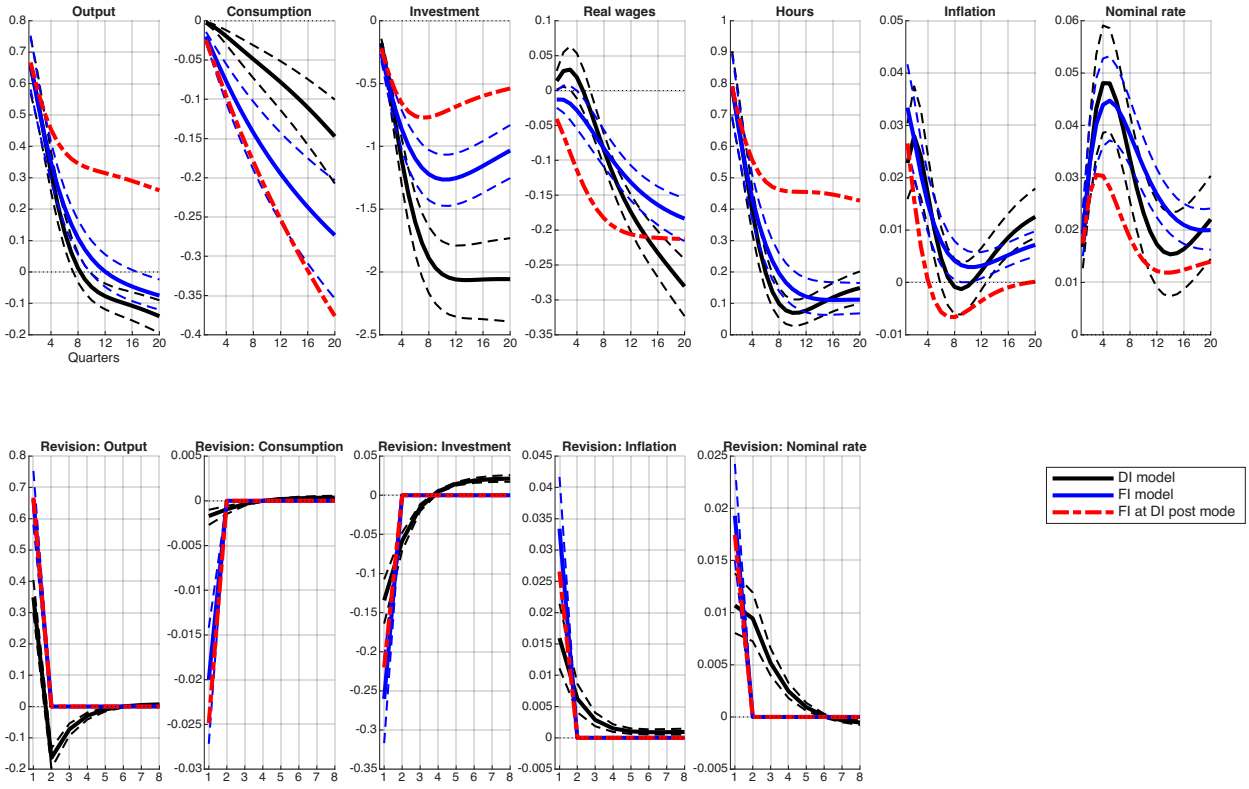
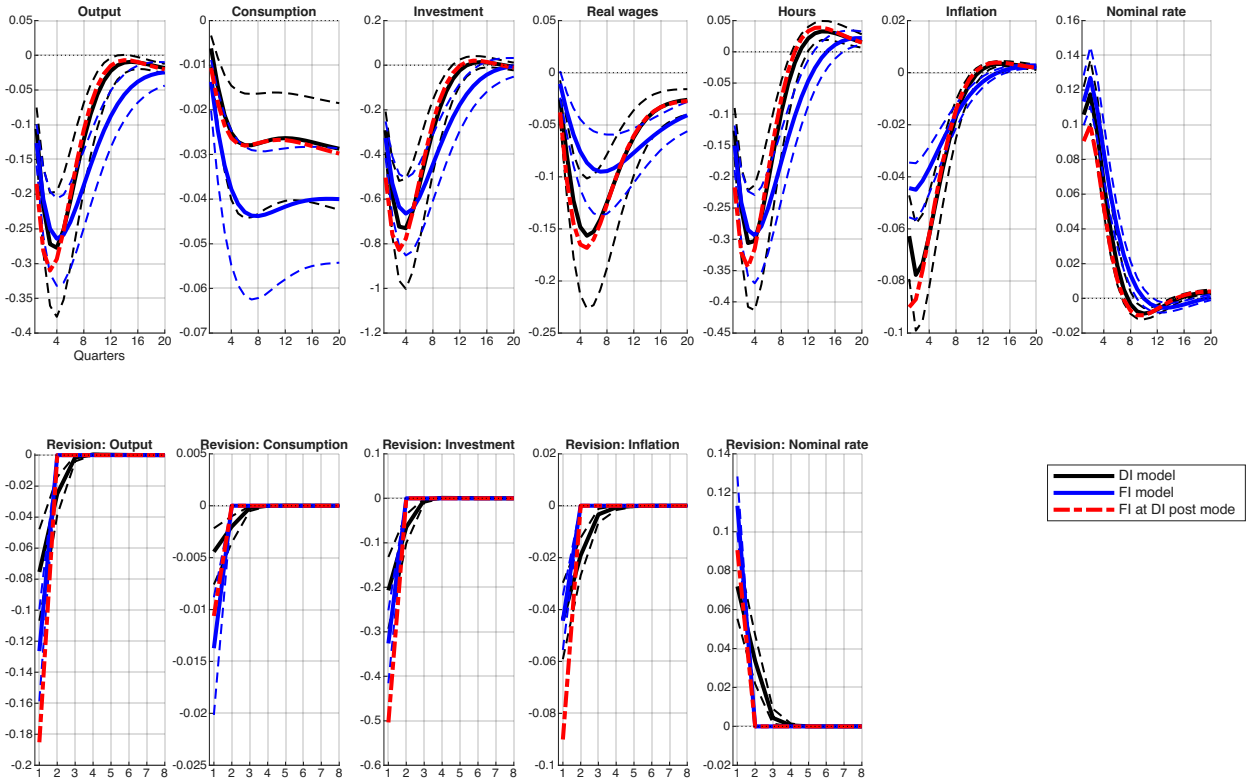


Figure J6: Impulse responses to monetary policy shock



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